

Brain Natriuretic Peptide Testing - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses brain natriuretic peptide testing.

1. Medical Necessity

Aetna considers measurement of plasma brain natriuretic peptide (BNP) medically necessary for the following indications:

1. To differentiate dyspnea due to heart failure from pulmonary disease; *or*
2. To determine prognosis or disease severity in chronic heart failure; *or*
3. To determine risk of developing heart failure in persons with risk factors (cardiovascular diseases, chemotherapy, chronic kidney disease, diabetes, family history of heart failure, obesity, radiotherapy, sleep apnea, and smoking); *or*
4. To screen for heart failure annually in adults with diabetes mellitus; *or*
5. Measurement upon hospital admission to determine prognosis in persons with acutely decompensated heart failure.

2. Experimental, Investigational, or Unproven

1. Aetna considers serial measurements of plasma BNP and/or its inactive metabolite, N-terminal pro-BNP (NT-proBNP), experimental, investigational, or unproven for all other indications, including any of the following, because the clinical value of their measurements for these indications and the effectiveness of these approaches has not been established:
 1. As a biomarker for cerebral small vessel disease/vascular brain damage in hypertension; *or*
 2. As a biomarker for hypertensive disorders of pregnancy; *or*
 3. As a biomarker for subclinical brain damage; *or*
 4. As a cardiac biomarker for Friedreich ataxia; *or*

5. As a cardiovascular biomarker in healthy normal subjects; *or*
 6. As a prognostic biomarker for acute coronary syndrome; *or*
 7. As a prognostic biomarker for the presence of diastolic dysfunction related to anemia in persons with sickle cell disease (NT-proBNP only); *or*
 8. As a prognostic biomarker for the risk of incident of type 2 diabetes; *or*
 9. As a prognostic biomarker of weaning outcome from mechanical ventilation; *or*
 10. As a prognostic marker for individuals with structural congenital heart disease; *or*
 11. For detecting early cardiac dysfunction in individuals with chronic fatigue syndrome; *or*
 12. For detecting early cardiac dysfunction in individuals with tetralogy of Fallot; *or*
 13. For determining prognosis of members after an acute coronary syndrome episode; *or*
 14. For diagnosing cardio-embolic stroke; *or*
 15. For diagnosing Kawasaki disease; *or*
 16. For diagnosing patent ductus arteriosus; *or*
 17. For diagnosing preeclampsia; *or*
 18. For diagnosing systemic sclerosis heart involvement; *or*
 19. For diagnosing or screening of pulmonary hypertension associated with bronchopulmonary dysplasia; *or*
 20. For guiding statin decisions for members with heart failure; *or*
 21. For guiding the initiation of thrombolytic therapy in members with acute pulmonary embolism; *or*
 22. For identifying individuals at risk of developing abnormal brain aging; *or*
 23. For identifying stress-induced myocardial ischemia; *or*
 24. For managing (diagnostic, prognostic and therapeutic) members with chronic renal failure; *or*
 25. For monitoring the effectiveness of therapy for members with congestive heart failure; *or*
 26. For prediction of acute kidney injury after non-cardiac surgery; *or*
 27. For prediction of cardiovascular complications after bariatric surgery; *or*
 28. For prediction of fatal outcome after stroke; *or*
 29. For prediction of outcomes after bi-ventricular repair in individuals with borderline hypoplastic left ventricle; *or*
 30. For prediction of outcome in congenital diaphragmatic hernia; *or*
 31. For prediction of short-term mortality in individuals with sepsis; *or*
 32. For prediction of the occurrence of atrial fibrillation after cryptogenic stroke or after thoracic surgery; *or*
 33. For pre-operative cardiac risk assessment in non-cardiac surgery (e.g., predicting adverse cardiovascular outcomes following non-cardiac surgery); *or*
 34. For routine evaluation of dyspnea, other than where necessary to distinguish heart failure from pulmonary disease; *or*
 35. For risk stratification of individuals with aortic stenosis; *or*
 36. For screening unrecognized left ventricular dysfunction; *or*
 37. For titrating therapy for members with chronic heart failure.
2. Aetna considers NT-proBNP testing experimental, investigational, or unproven for for screening left ventricular systolic dysfunction in asymptomatic long-term breast cancer

survivors because the effectiveness of this approach has not been established.

3. Aetna considers the use of serum NT-proBNP, alone or in combination with D-dimer, albumin combined with T-cell subsets, experimental, investigational, or unproven for detecting coronary artery damage in children with Kawasaki disease because the effectiveness of this approach has not been established.

3. Related Policies

- [CPB 0709 - Nesiritide \(Natreacor\)](#)

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
83880	Natriuretic peptide [except as a cardiovascular biomarker in healthy normal subjects and for identifying stress-induced myocardial ischemia]
Other CPT codes related to the CPB:	
82040	Albumin; serum, plasma or whole blood
82042	Albumin; other source, quantitative, each specimen
82043	Albumin; urine (eg, microalbumin), quantitative
82044	Albumin; urine (eg, microalbumin), semiquantitative (eg, reagent strip assay)
82045	Albumin; ischemia modified
85378	Fibrin degradation products, D-dimer; qualitative or semiquantitative
85379	Fibrin degradation products, D-dimer; quantitative
85380	Fibrin degradation products, D-dimer; ultrasensitive (eg, for evaluation for venous thromboembolism), qualitative or semiquantitative
86359	T cells; total count
86360	T cells; absolute CD4 and CD8 count, including ratio
86361	T cells; absolute CD4 count
93306	Echocardiography, transthoracic, real-time with image documentation (2D), includes M-mode recording, when performed, complete, with spectral Doppler echocardiography, and with color flow Doppler echocardiography
93307	Echocardiography, transthoracic, real-time with image documentation (2D), includes M-mode recording, when performed, complete, without spectral or color Doppler echocardiography
93308	Echocardiography, transthoracic, real-time with image documentation (2D), includes M-mode recording, when performed, follow-up or limited study
93312	Echocardiography, transesophageal, real-time with image documentation (2D) (with or without M-mode recording); including probe placement, image acquisition, interpretation and report

Code	Code Description
93313	Echocardiography, transesophageal, real-time with image documentation (2D) (with or without M-mode recording); placement of transesophageal probe only
93314	Echocardiography, transesophageal, real-time with image documentation (2D) (with or without M-mode recording); image acquisition, interpretation and report only
ICD-10 codes covered if selection criteria are met:	
C00.0-D49.9	Neoplasms
E08.00-E13.9	Diabetes mellitus
• E66.01-E66.2, • E66.811-E66.813, • E66.89, • E66.9	Obesity
F17.200-F17.219	Nicotine dependence [smoking]
F17.290-F17.299	Nicotine dependence, other tobacco product [pipes, cigars]
G47.30-G47.39	Sleep apnea
• I00-I05.9, • I06.1-I11.9, • I15.0-I1A.0, • I30.0-I34.89, • I35.1, • I35.8-I47.9, • I48.3-I49.9, • I52-I63.9, • I68.0-I99.9	Cardiovascular diseases
I06.0	Rheumatic aortic stenosis [not covered for risk stratification of individuals with aortic stenosis]
I12.0 - I12.9	Hypertensive chronic kidney disease [not covered for managing (diagnostic, prognostic and therapeutic) members with chronic renal failure]

Code	Code Description
I13.0 - I13.2	Hypertensive heart and chronic kidney disease [not covered for managing (diagnostic, prognostic and therapeutic) members with chronic renal failure]
I20.0 - I25.9	Ischemic heart diseases [not covered for identification of stress-induced myocardial ischemia][including cardio-embolic stroke]
I26.01 - I28.9	Pulmonary heart disease and diseases of pulmonary circulation [not covered for guiding the initiation of thrombolytic therapy in members with acute pulmonary embolism]
I35.0	Nonrheumatic aortic (valve) stenosis [not covered for risk stratification of individuals with aortic stenosis]
I35.2	Nonrheumatic aortic (valve) stenosis with insufficiency [not covered for risk stratification of individuals with aortic stenosis]
I48.0 - I48.21	Atrial fibrillation [not covered for prediction of atrial fibrillation after cryptogenic stroke] [not covered for the prediction of atrial fibrillation after thoracic surgery]
I50.1 - I50.9	Heart failure
I51.0 - I51.9	Complications and ill-defined descriptions of heart disease [not covered for screening unrecognized left ventricular dysfunction]
I65.01 - I67.9	Occlusion and stenosis of precerebral and cerebral arteries and transient cerebral ischemia [stroke or transient ischemic attack of obscure or unknown origin] [not covered for prediction of fatal outcome after stroke] [including cardio-embolic stroke]
J40 - J47.9	Chronic lower respiratory diseases
N18.1-N18.9	Chronic kidney disease (CKD) [not covered for managing (diagnostic, prognostic and therapeutic) members with chronic renal failure]
R06.00 - R06.09	Dyspnea [not covered for routine evaluation of dyspnea, other than where necessary to distinguish heart failure from pulmonary disease]
R06.82	Tachypnea, not elsewhere classified [not covered for routine evaluation of dyspnea, other than where necessary to distinguish heart failure from pulmonary disease]
Z51.0	Encounter for antineoplastic radiation therapy
Z51.11- Z51.12	Encounter for antineoplastic chemotherapy and immunotherapy
Z68.30- Z68.45	Body mass index [BMI]
Z68.54- Z68.56	Body mass index [BMI] pediatric
Z72.0	Tobacco use [smoking]
Z87.891	Personal history of nicotine dependence
Z79.630- Z79.634	Long term (current) use of chemotherapeutic agent
Z82.49	Family history of ischemic heart disease and other diseases of the circulatory system
Z92.3	Personal history of irradiation

Code	Code Description
ICD-10 codes not covered for indications listed in the CPB:	
• A02.1, • A22.7, • A24.1, • A26.7, • A31.2, • A32.7, • A40.0 - A40.9, • A41.01 - A41.9, • A42.7, • A54.86, • B00.7, • B37.7, • O03.37, • O03.87, • O04.87, • O07.37, • O08.82, • O75.3, • O85, • O86.04, • P36.0 - P36.9 , • R65.20 - R65.21, • T81.44xA - T81.44xS, • T88.0xxA - T88.0xxS	Sepsis
D57.00 - D57.819	Sickle-cell disorders [sickle-cell anemia]
F01.50 - F01.C4	Vascular dementia
G11.11 - G11.19	Early-onset cerebellar ataxia
G45.0 - G45.9	Transient cerebral ischemic attacks and related syndromes

Code	Code Description
G93.0 - G93.9	Other disorders of brain [subclinical brain damage]
J86.0 - J86.9	Pyothorax [not covered for guiding the initiation of thrombolytic therapy in members with acute pulmonary embolism]
M30.3	Mucocutaneous lymph node syndrome [Kawasaki] [not covered for diagnosing Kawasaki disease]
M34.89	Other systemic sclerosis [Heart involvement]
011.1 - 011.9	Pre-existing hypertension with pre-eclampsia
013.1 - 013.9	Gestational [pregnancy-induced] hypertension without significant proteinuria
014.00 - 014.95	Pre-eclampsia
P27.1	Bronchopulmonary dysplasia originating in the perinatal period
Q20.0 - Q28.9	Congenital malformations of the circulatory system [not covered for detection of early cardiac dysfunction in individuals with tetralogy of Fallot] [not covered for risk stratification of individuals with aortic stenosis]
Q79.0	Congenital diaphragmatic hernia [not covered for the prediction of outcome in congenital diaphragmatic hernia]
R41.81	Age-related cognitive decline [not covered for identifying individuals at risk of developing abnormal brain aging]
R53.82	Chronic fatigue, unspecified
• S06.0X0A - S06.A1XS, • S06.0XAA - S06.9XAS	Intracranial injury [subclinical brain damage]
Z01.810	Encounter for preprocedural cardiovascular examination [pre-operative cardiac risk assessment in non-cardiac surgery]
Z13.6	Encounter for screening for cardiovascular disorders [pre-operative cardiac risk assessment in non-cardiac surgery][Screening left ventricular systolic dysfunction]
Z13.89	Encounter for screening for other disorder
Z85.3	Personal history of malignant neoplasm of breast
Z99.11	Dependence on respirator [ventilator] status

Background

This policy is based in part upon the 2017 American College of Cardiology/American Heart Association/Heart Failure Society of America (ACC/AHA/HFSA) Focused Update of the ACCF/AHA 2013 Guideline for the Management of Heart Failure.

Plasma brain natriuretic peptide (BNP) is a 32-amino acid polypeptide that contains a 17-amino acid ring structure common to all natriuretic peptides. The cardiac ventricles are the major source of plasma BNP. This circulating peptide has been used as a marker to assist in the diagnosis of congestive heart failure. In general, plasma BNP levels correlate positively with the degree of left ventricular dysfunction, but they are

sensitive to other biological factors such as age, sex, and diastolic dysfunction. Plasma BNP levels greater than 100 pg/ml are reported to support a diagnosis of abnormal or symptomatic heart failure.

Guidelines on heart failure from the American College of Cardiology, the American Heart Association and the Heart Failure Society of America (Yancey, et al., 2017) includes the following strong recommendations based on high-quality evidence:

- Natriuretic peptide biomarkers should be measured in patients presenting with dyspnea to help diagnose or exclude heart failure.
- B-type natriuretic peptide or N-terminal pro-B-type natriuretic peptide should be measured to determine prognosis or disease severity in chronic heart failure.
- Baseline natriuretic peptide biomarkers, cardiac troponin, or both should be measured upon hospital admission to determine prognosis in patients with acutely decompensated heart failure.

The ACC/AHA practice guidelines on heart failure (Hunt et al, 2005) stated the following conclusions about the clinical utility of BNP: "Measurement of B-type natriuretic peptide (BNP) can be useful in the evaluation of patients presenting in the urgent care setting in whom the clinical diagnosis of heart failure (HF) is uncertain. (Level of Evidence: A)."

The guidelines stated, however, that "[t]he value of serial measurements of BNP to guide therapy for patients with HF is not well established. (Level of Evidence: C)."

The guidelines explained that serum BNP levels have been shown to parallel the clinical severity of heart failure in broad populations (Hunt et al, 2005). Levels are higher in hospitalized patients and tend to decrease during aggressive therapy for decompensation. The guidelines stated, however, that it can not be assumed that BNP levels can be used effectively as targets for adjustment of therapy in individual patients. The guidelines explained that many patients taking optimal doses of medications continue to show markedly elevated levels of BNP, and some patients demonstrated BNP levels within the normal range despite advanced heart failure (HF). The guidelines concluded that the use of BNP measurements to guide the titration of drug doses has not been shown to improve outcomes more effectively than achievement of the target doses drugs shown in clinical trials to prolong life. The guidelines noted that ongoing trials will help to determine the role of serial BNP measurements in both diagnosis and management of heart failure.

Regarding the use of BNP to assess prognosis, the guidelines stated that elevated BNP levels predict higher risk of heart failure and other events after myocardial infarction, whereas marked elevation in BNP levels during hospitalization for heart failure may predict re-hospitalization and death (Hunt et al, 2005). The guidelines concluded, however, that "the BNP measurement has not been clearly shown to supplement careful clinical assessment."

Thus, measurement of plasma BNP may be medically necessary to differentiate dyspnea due to heart failure from pulmonary disease in the urgent care setting. The value of measurements of BNP for the routine (non-urgent) diagnosis or for the management of patients with heart failure has not been established.

A technology assessment of BNP for the diagnosis and management of congestive heart failure by the Institute for Clinical Systems Improvement (2005) stated that "BNP testing is useful as an adjunct to other clinical tools for differentiating cardiac (congestive heart failure [CHF]) causes from other causes of dyspnea presenting in the emergency department or urgent care setting." The ICSI technology assessment stated that, in particular, the diagnosis of CHF is highly unlikely in patients with normal BNP levels. The ICSI technology assessment states that care should be taken when measuring BNP within 2 to 4 hours after the onset of acute symptoms as false negatives may occur. The ICSI technology assessment concluded that there are no data to support the use of BNP in the general screening of asymptomatic populations for CHF, and thus BNP testing should not be used for this purpose. The ICSI technology assessment also concluded that the utility of BNP as a tool to optimize management of heart failure or measure treatment response has yet to be defined. "Serial testing of BNP levels has not been shown to have clinical utility" (ICSI, 2005).

In a review on the use of BNP as a potential marker of acute coronary syndromes, Body and Roberts (2006) stated that the clinical bottom line is that BNP shows promise as an early

cardiac marker and may enhance prognostic stratification. Negative-predictive value and positive-predictive value may be unacceptably low to enable use as a sole cardiac marker. Incorporation into a multi-marker strategy and serial estimations may be necessary.

Sohne and associates (2006) determined the predictive value of elevated BNP levels for early recurrent venous thromboembolism with or without fatal outcome in hemodynamically stable patients with acute pulmonary embolism (PE). In addition, these researchers evaluated the potential clinical consequences of initiating thrombolytic therapy based on the BNP levels alone. A nested case-control study was performed within the framework of a large randomized-controlled trial totaling 2,213 hemodynamically stable patients with confirmed acute, symptomatic PE. A total of 90 patients experienced a fatal or non-fatal recurrent venous thromboembolism during the first 3 months of follow-up (cases); 297 patients with uneventful follow-up served as controls. Blood for BNP levels was obtained at referral and assayed in a central laboratory. Cases had significantly higher mean baseline BNP levels ($p = 0.0002$). The odds ratio (OR) for every logarithmic unit increase in BNP concentration was 2.4 (95 % confidence interval [CI]: 1.5 to 3.7). A BNP cut-off level of 1.25 pmol/L [the optimal point on the receiver-operating characteristic curve] was associated with a sensitivity and specificity of 60 % and 62 %, respectively. In theory, for every patient correctly receiving thrombolytic therapy at this cut-off, 16 patients will receive this therapy unnecessarily. These investigators concluded that BNP level at presentation is significantly associated with early (fatal) recurrent venous thromboembolism in hemodynamically stable patients with acute PE. However, this relationship appears clinically insufficient to guide the initiation of thrombolytic therapy.

The Agency for Healthcare Research and Quality's assessment on testing for BNP and the N-terminal fragment of B-type natriuretic peptide (NT-proBNP) in the diagnosis and prognosis of heart failure (Balion et al, 2006) stated that these natriuretic peptides can be used to rule out heart failure in patients being seen in emergency rooms, specialized clinics, and primary care settings. It also noted that there were few studies that examined B-type natriuretic peptides in populations without known heart failure. All but a single study suggested that measurements of these biomarkers are inaccurate to be an effective screening test for unrecognized left ventricular dysfunction.

Although several studies have addressed the use of biomarkers -- particularly BNP and NT-proBNP -- in populations with heart failure (HF), integrating these markers into clinical care has been controversial. The National Academy of Clinical Biochemistry (NACB) convened a committee to develop practice guidelines for the use of biomarkers for screening, diagnosis, prognostication, and treatment of HF (Tang et al, 2007). Some of the key points of this practice guideline are as follows:

- Although natriuretic peptide levels, including longitudinal measurements, may be useful for additional risk stratification in some patients, routine use solely for HF risk stratification is discouraged (Class III recommendation).
- Natriuretic peptide levels may be influenced by several patient factors, including age, sex, renal function, thyroid function, anemia, and body habitus. Importantly, obese persons tend to have lower natriuretic peptide levels than do non-obese persons.
- Natriuretic peptide levels should not replace standard clinical assessment tools, such as echocardiography (Class III recommendation)
- Normal BNP and NT-proBNP ranges vary according to the assay used and the characteristics of the control population. The assay commonly used for research produces systematically lower measurements than do commercial assays.
- The committee made only one Class I recommendation for the clinical use of natriuretic peptides: to exclude or confirm the diagnosis of HF in patients with ambiguous signs and symptoms in the acute setting. Such an application in the non-acute setting received a Class IIa recommendation for lack of studies.
- The routine use of natriuretic peptides in the initial evaluation of patients with suspected HF, for guiding therapy in patients with established HF, and for screening purposes is also discouraged (Class III recommendations).

Thus, available data support relatively few strong recommendations for the clinical use of natriuretic peptide measurements in patients with HF, other than adjunctive use for

diagnosis in the acute care setting. Until more evidence is available on how these cardiac biomarkers should be integrated into clinical care, their routine use in the diagnosis, treatment, and screening of HF is not warranted.

Guidelines from the American College of Cardiology (Heidenreich, et al., 2022) state that "Measurement of BNP and NT-proBNP levels in the ambulatory setting for a suspected cardiac cause of dyspnea provides incremental diagnostic value to clinical judgment when the cause of dyspnea is unclear and the physical examination equivocal." The guidelines state that, regarding use in directing therapy in hospitalized heart failure patients: "Predischarge BNP and NT-proBNP levels are strong predictors of the risk of death or hospital readmission for HF. Although patients in whom levels of BNP or NT-proBNP decreased with treatment had better outcomes than those without any changes or with a biomarker rise, targeting a certain threshold, value, or relative change in these biomarker levels during hospitalization has not been shown to be consistently effective in improving outcomes. Patients in which GDMT leads to a reduction in BNP and NT-proBNP levels represent a population with improved long-term outcomes compared with those with persistently elevated levels despite appropriate treatment. BNP and NT-proBNP levels and their change could help guide discussions on prognosis as well as adherence to, and optimization of, GDMT." Regarding repeat testing for directing therapy of heart failure patients, the guidelines state, however, that "[a]lthough a reduction in BNP and NT-proBNP has been associated with better outcomes, the evidence for treatment guidance using serial BNP or NT-proBNP measurements remains insufficient."

Rottlaender et al (2008) stated that several factors (e.g., age, sex, obesity as well as chronic renal failure) have to be considered in the interpretation of natriuretic peptides, which may support diagnostics of HF in patients with unexplained dyspnea. However, cardiac biomarkers should not be used to replace conventional clinical evaluation. The use of natriuretic peptides for screening asymptomatic populations is inappropriate. A BNP-guided titration of HF medication is not yet warranted. Brain natriuretic peptide testing may be used only in selected situations for risk stratification since the prognostic value is still limited by a lack of clear usefulness in guiding clinical management. The authors concluded that measurements of natriuretic peptides are at present largely an addition in the diagnosis of acute HF, as long as possible errors in interpretation are taken into account.

Mark and colleagues (2007) stated that premature cardiovascular disease is the leading cause of morbidity and mortality in patients with end-stage renal failure. Natriuretic peptides, specifically BNP, are released from the heart in response to chamber distension and thus increased in the presence of volume expansion and cardiac overload. Their physiological role is to cause vasodilatation and promote natriuresis to maintain volume homeostasis. However, the diagnostic role of serum BNP levels in patients with advanced renal dysfunction remains to be defined. This is in agreement with the observation of Rosner (2007) who noted that the diagnostic utility of BNP in end-stage renal disease is limited.

Pfisterer et al (2009) stated that it is unclear if intensified HF therapy guided by N-terminal BNP is superior to symptom-guided therapy. In a randomized, controlled multicenter study, these investigators compared 18-month outcomes of N-terminal BNP-guided versus symptom-guided HF therapy. A total of 499 patients aged 60 years or older with systolic HF (ejection fraction less than or equal to 45 %), New York Heart Association (NYHA) class of II or greater, prior hospitalization for HF within 1 year, and N-terminal BNP level of 2 or more times the upper limit of normal were included in this trial. The study had an 18-month follow-up and was conducted at 15 outpatient centers in Switzerland and Germany. Interventions were up-titration of guideline-based treatments to reduce symptoms to NYHA class of II or less (symptom-guided therapy) and BNP level of 2 times or less the upper limit of normal and symptoms to NYHA class of II or less (BNP-guided therapy). Primary outcomes were 18-month survival free of all-cause hospitalizations and quality of life as assessed by structured validated questionnaires. Heart failure therapy guided by N-terminal BNP and symptom-guided therapy resulted in similar rates of survival free of all-cause hospitalizations (41 % versus 40 %, respectively; hazard ratio [HR], 0.91 [95 % CI: 0.72 to 1.14]; $p = 0.39$). Patients' quality-of-life metrics improved over 18 months of follow-up, but these improvements were similar in both the N-terminal BNP-guided and symptom-guided strategies. Compared with the symptom-guided group, survival free of hospitalization for HF, a secondary end point, was higher among those in the N-terminal BNP-guided group (72 % versus 62 %, respectively; HR, 0.68 [95 % CI: 0.50 to 0.92]; $p = 0.01$). Heart failure therapy guided by N-terminal BNP improved outcomes in patients aged 60 to 75 years but not in those aged 75 years or older ($p <$

0.02 for interaction). The authors concluded that HF therapy guided by N-terminal BNP did not improve overall clinical outcomes or quality of life compared with symptom-guided treatment.

Schneider et al (2009) noted that BNP is used to diagnose HF, but the effects of using the test on all dyspneic patients is uncertain. In a randomized, single-blind trial, these researchers evaluated if BNP testing alters clinical outcomes and health services use of acutely dyspneic patients. Patients were blinded to the intervention, but clinicians and those who assessed trial outcomes were not. A total of 612 consecutive patients who presented with acute severe dyspnea were included in this study (n = 306 for BNP testing; n = 306 for no testing). Primary outcome measures included admission rates, length of stay, and emergency department medications; secondary outcomes were mortality and re-admission rates. There were no between-group differences in hospital admission rates (85.6 % [BNP group] versus 86.6 % [control group]; difference, -1.0 percentage point [95 % CI: -6.5 to 4.5 percentage points]; p = 0.73), length of admission (median of 4.4 days [inter-quartile range, 2 to 9 days] versus 5.0 days [inter-quartile range of 2 to 9 days]; p = 0.94), or management of patients in the emergency department. Test discrimination was good (area under the receiver-operating characteristic curve, 0.87 [CI: 0.83 to 0.91]). Adverse events were not measured. The limitations of this study were that most patients were very short of breath and required hospitalization; the findings might not apply for evaluating patients with milder degrees of breathlessness. The authors concluded that measurement of BNP in all emergency department patients with severe shortness of breath had no apparent effects on clinical outcomes or use of health services. It does not improve admission or discharge decisions or improve initial treatment planning. The findings do not support routine use of BNP testing in all severely dyspneic patients in the emergency department.

Karthikeyan and colleagues (2009) performed a systematic review and meta-analysis to determine if pre-operative BNP (i.e., BNP or N-terminal pro-B-type natriuretic peptide [NT-proBNP]) is an independent predictor of 30-day adverse cardiovascular outcomes after non-cardiac surgery. These investigators employed 5 search strategies (e.g., searching bibliographic databases), and included all studies that assessed the independent prognostic value of pre-operative BNP measurement as a predictor of cardiovascular complications after non-cardiac surgery. These researchers determined study eligibility and conducted data abstraction independently and in duplicate. They calculated a pooled odds ratio using a random effects model. A total of 9 studies met eligibility criteria, and included a total of 3,281 patients, among whom 314 experienced 1 or more peri-operative cardiovascular complications. The average proportion of patients with elevated BNP was 24.8 % (95 % CI: 20.1 % to 30.4 %; I(2) = 89 %). All studies showed a statistically significant association between an elevated pre-operative BNP level and various cardiovascular outcomes (e.g., a composite of cardiac death and non-fatal myocardial infarction; atrial fibrillation). Data pooled from 7 studies demonstrated an odds ratio (OR) of 19.3 (95 % CI: 8.5 to 43.7; I(2) = 58 %). The pre-operative BNP measurement was an independent predictor of peri-operative cardiovascular events among studies that only considered the outcomes of death, cardiovascular death, or myocardial infarction (OR: 44.2, 95 % CI: 7.6 to 257.0, I(2) = 51.6 %), and those that included other outcomes (OR: 14.7, 95 % CI: 5.7 to 38.2, I(2) = 62.2 %); the p value for interaction was 0.28. The authors concluded that these results suggested that an elevated pre-operative BNP or NT-proBNP measurement is a powerful, independent predictor of cardiovascular events in the first 30 days after non-cardiac surgery.

In an editorial that accompanied the afore-mentioned paper, Bolliger et al (2009) stated that the study by Karthikeyan et al provided evidence for a high prognostic potential of NPs in patients scheduled for non-cardiac surgery. However, studies to evaluate if specific NP-based treatment modifications will result in improved outcome of surgical patients still need to be performed. Should future studies find outcome relevance of such a concept, NPs will be indeed the magic bullet of pre-operative risk optimization. So far, however, they are interesting and promising tools for risk stratification that requires further evaluation.

Cleland et al (2009) examined if plasma NT-proBNP a marker of cardiac dysfunction and prognosis measured in CORONA (Controlled Rosuvastatin Multinational Trial in Heart Failure) could be used to identify the severity of HF at which statins become ineffective. In CORONA, patients with HF, reduced left ventricular ejection fraction, and ischemic heart disease were randomly assigned to 10 mg/day rosuvastatin or placebo. The primary composite outcome was cardiovascular death, non-fatal myocardial infarction, or stroke. Of 5,011 patients enrolled, NT-proBNP was measured in 3,664 (73 %). The mid-

tertile included values between 103 pmol/L (868 pg/ml) and 277 pmol/L (2,348 pg/ml). Log NT-proBNP was the strongest predictor (per log unit) of every outcome assessed but was strongest for death from worsening HF (HR: 1.99; 95 % CI: 1.71 to 2.30), was weaker for sudden death (HR: 1.69; 95 % CI: 1.52 to 1.88), and was weakest for athero-thrombotic events (HR: 1.24; 95 % CI: 1.10 to 1.40). Patients in the lowest tertile of NT-proBNP had the best prognosis and, if assigned to rosuvastatin rather than placebo, had a greater reduction in the primary end point (HR: 0.65; 95 % CI: 0.47 to 0.88) than patients in the other tertiles (heterogeneity test, $p = 0.0192$). This reflected fewer athero-thrombotic events and sudden deaths with rosuvastatin. The authors concluded that patients with HF due to ischemic heart disease who have NT-proBNP values less than 103 pmol/l (868 pg/ml) may benefit from rosuvastatin.

In an editorial that accompanied the study by Cleland et al, Daniels and Barrett-Connor (2009) stated that clinical practice guidelines recommend that statins be prescribed to patients with ischemic heart disease, but do not make HF a consideration. If these findings are confirmed in other studies, NP levels may have a new application in guiding statin decisions for HF patients.

In a meta-analysis, Porapakkham and associates (2010) examined the overall effect of BNP-guided drug therapy on cardiovascular outcomes in patients with chronic HF. These researchers identified randomized controlled trials (RCTs) by systematic search of manuscripts, abstracts, and databases. Eligible RCTs were those that enrolled more than 20 patients and involved comparison of BNP-guided drug therapy versus usual clinical care of the patient with chronic HF in an out-patient setting. Eight RCTs with a total of 1,726 patients and with a mean duration of 16 months (range of 3 to 24 months) were included in the meta-analysis. Overall, there was a significantly lower risk of all-cause mortality (relative risk [RR], 0.76; 95 % CI: 0.63 to 0.91; $p = 0.003$) in the BNP-guided therapy group compared with the control group. In the subgroup of patients younger than 75 years, all-cause mortality was also significantly lower in the BNP-guided group (RR, 0.52; 95 % CI: 0.33 to 0.82; $p = 0.005$). However, there was no reduction in mortality with BNP-guided therapy in patients 75 years or older (RR, 0.94; 95 % CI: 0.71 to 1.25; $p = 0.70$). The risk of all-cause hospitalization and survival free of any hospitalization was not significantly different between groups (RR, 0.82; 95 % CI: 0.64 to 1.05; $p = 0.12$ and RR, 1.07; 95 % CI: 0.85 to 1.34; $p = 0.58$, respectively). The additional percentage of patients achieving target doses of angiotensin-converting enzyme inhibitors and beta-blockers during the course of these trials averaged 21 % and 22 % in the BNP group and 11.7 % and 12.5 % in the control group, respectively. The authors concluded that B-type natriuretic peptide-guided therapy reduces all-cause mortality in patients with chronic HF compared with usual clinical care, especially in patients younger than 75 years. A component of this survival benefit may be due to increased use of agents proven to decrease mortality in chronic HF. However, there does not seem to be a reduction in all-cause hospitalization or an increase in survival free of hospitalization using this approach.

Eurlings et al (2010) examined if management of HF guided by an individualized NT-proBNP target would lead to improved outcome compared with HF management guided by clinical assessment alone. A total of 345 patients hospitalized for decompensated, symptomatic HF with elevated NT-proBNP levels at admission were included. After discharge, patients were randomized to either clinically-guided outpatient management ($n = 171$), or management guided by an individually set NT-proBNP ($n = 174$) defined by the lowest level at discharge or 2 weeks thereafter. The primary end point was defined as number of days alive outside the hospital after index admission. Management of HF guided by this individualized NT-proBNP target increased the use of HF medication ($p = 0.006$), and 64 % of HF-related events were preceded by an increase in NT-proBNP. Nevertheless, HF management guided by this individualized NT-proBNP target did not significantly improve the primary end point (685 versus 664 days, $p = 0.49$), nor did it significantly improve any of the secondary end points. In the NT-proBNP-guided group mortality was lower, as 46 patients died (26.5 %) versus 57 (33.3 %) in the clinically-guided group, but this was not statistically significant ($p = 0.206$). The authors concluded that serial NT-proBNP measurement and targeting to an individual NT-proBNP value did result in advanced detection of HF-related events and importantly influenced HF-therapy, but failed to provide significant clinical improvement in terms of mortality and morbidity.

In an editorial that accompanied the afore-mentioned study by Eurlings et al, Troughton et al (2010) stated that "further data are needed from more robust, adequately powered trials with hard clinical outcomes and from a meta-analysis utilizing individual patient data (rather than summary grouped data) before guidelines can confidently endorse a

biomarker-guided strategy ... Whether the biomarker-guided strategy is applicable to elderly patients and those with heart failure and preserved left ventricular ejection fraction remains unclear and needs further evaluation". Furthermore, Kim and Januzzi (2011) noted that "although evidence is increasing that NP-guided outpatient management of HF may improve clinical outcomes, more information is needed before adoption of such an approach, which is currently being tested in clinical trials".

Previous studies reported that plasma NT-proBNP has prognostic value for cardiovascular events in the general population even in the absence of HF. It is unclear if NT-proBNP retains predictive value in healthy normal subjects. McKie and associates (2010) determined the prognostic value of plasma NT-proBNP for death and cardiovascular events among subjects without risk factors for HF, which the authors termed healthy normal. These investigators identified a community-based cohort of 2,042 subjects in Olmsted County, Minnesota. Subjects with symptomatic (stage C/D) HF were excluded. The remaining 1,991 subjects underwent echocardiography and NT-proBNP measurement. These researchers further defined healthy normal ($n = 703$) and stage A/B HF ($n = 1,288$) subgroups. Healthy normal was defined as the absence of traditional clinical cardiovascular risk factors and echocardiographic structural cardiac abnormalities. Subjects were followed for death, HF, cerebrovascular accident, and myocardial infarction with median follow-up of 9.1, 8.7, 8.8, and 8.9 years, respectively. NT-proBNP was not predictive of death or cardiovascular events in the healthy normal subgroup. Similar to previous reports, in stage A/B HF, plasma NT-proBNP values greater than age-/sex-specific 80th percentiles were associated with increased risk of death, HF, cerebrovascular accident, and myocardial infarction ($p < 0.001$ for all) even after adjustment for clinical risk factors and structural cardiac abnormalities. The authors concluded that these findings do not support the use of NT-proBNP as a cardiovascular biomarker in healthy normal subjects.

Nadir and colleagues (2011) noted that studies in victims of sudden cardiac death and those surviving a cardiac arrest have confirmed that extent of coronary artery disease is similar in those with and without angina, suggesting that it is the presence of myocardial ischemia rather than associated symptoms that determine the prognosis. Experimental models show that hypoxic myocardial tissue results in production of extra BNP, suggesting that BNP could potentially serve as a biomarker of myocardial ischemia. These investigators performed a meta-analysis of the studies that link BNP to inducible myocardial ischemia as indicated by non-invasive stress tests. Values of true-positive, false-positive, true-negative, and false-negative were calculated from the reported sensitivity, specificity, disease prevalence, and total number of patients studied. A total of 16 studies reporting data on 2,784 patients across 14 study populations were included in the final analysis. Mean age of participants was 55 to 69 years and 55 % to 90 % were men. Pooled sensitivity and specificity of BNP for detection of stress-induced myocardial ischemia were 71 % (95 % CI: 68 to 74) and 52 % (95 % CI: 52 to 54), respectively. Pooled diagnostic odds ratio was 3.5 (95 % CI: 2.46 to 5.04) and summary receiver operating characteristic curve revealed an area under the curve of 0.71 ± 0.02 (mean $\pm$ SE). The authors concluded that this meta-analysis suggests that an increased BNP level can identify inducible ischemia as detected by standard non-invasive stress tests. They stated that this raises the possibility of a whole new role for BNP in the diagnosis and management of myocardial ischemia.

Pfister et al (2011) noted that genetic and epidemiological evidence suggests an inverse association between BNP levels in blood and risk of type 2 diabetes (T2D), but the prospective association of BNP with T2D is uncertain, and it is unclear whether the association is confounded. In a prospective, case-cohort study, these researchers analyzed the association between levels of the NT-proBNP in blood and risk of incident T2D and genotyped the variant rs198389 within the BNP locus in 3 T2D case-control studies. They combined their results with existing data in a meta-analysis of 11 case-control studies. Using a Mendelian randomization approach, these investigators compared the observed association between rs198389 and T2D to that expected from the NT-proBNP level to T2D association and the NT-proBNP difference per C allele of rs198389. In participants of this case-cohort study who were free of T2D and cardiovascular disease at baseline, these researchers observed a 21 % (95 % CI: 3 % to 36 %) decreased risk of incident T2D per 1 standard deviation (SD) higher log-transformed NT-proBNP levels in analysis adjusted for age, sex, body mass index, systolic blood pressure, smoking, family history of T2D, history of hypertension, and levels of triglycerides, high-density lipoprotein cholesterol, and low-density lipoprotein cholesterol. The association between rs198389 and T2D observed in case-control studies (odds ratio= 0.94 per C allele, 95 % CI: 0.91 to 0.97) was similar to that expected (0.96: 0.93 to 0.98) based on the

pooled estimate for the log-NT-proBNP level to T2D association derived from a meta-analysis of the authors' study and published data (hazard ratio = 0.82 per SD, 0.74 to 0.90) and the difference in NT-proBNP levels (0.22 SD, 0.15 to 0.29) per C allele of rs198389. No significant associations were observed between the rs198389 genotype and potential confounders. The authors concluded that these findings provided evidence for a potential causal role of the BNP system in the etiology of T2D. They stated that further studies are needed to investigate the mechanisms underlying this association and possibilities for preventive interventions.

In a single-center, retrospective study, Takatsuki et al (2012) examined if NT-proBNP was a biomarker of clinical, laboratory, and echocardiographic abnormalities in children with homozygous sickle cell disease. This study consisted of analysis of data from November 2007 to December 2010. These investigators correlated serum NT-proBNP with clinical and laboratory findings, echocardiographic data, and NYHA functional class. NT-proBNP levels from 42 children (median age of 9 years; 52 % female) had significant correlations with hemoglobin ($r = -0.63$, $p < 0.05$), and echocardiographic measurements including tricuspid regurgitant velocity ($r = 0.46$, $p < 0.05$), lateral E' ($r = -0.52$, $p < 0.05$), and lateral E/E' ratio (an indicator of left ventricular filling pressures and is used in the assessment of diastolic dysfunction) ($r = 0.60$, $p < 0.05$), suggesting diastolic dysfunction. In addition, NT-proBNP levels increased from NYHA functional class I to class III and had a significant linear correlation with the NYHA functional class ($r = 0.69$, $p < 0.05$). The authors concluded that NT-proBNP correlated with low hemoglobin and tissue Doppler data as indicators of diastolic dysfunction. Elevated NT-proBNP may be a prognostic biomarker for the presence of diastolic dysfunction related to anemia in children with sickle cell disease. The findings of this small, retrospective study need to be validated by well-designed studies.

Lin and colleagues (2012) noted that novel biomarkers of myocardial ischemia and inflammatory processes have the potential to improve diagnostic accuracy of acute coronary syndrome (ACS) within a shorter time interval after symptom onset. These researchers reviewed the recent literature and evaluated the evidence for use of novel biomarkers in diagnosing ACS in patients presenting with chest pain or symptoms suggestive of cardiac ischemia to the emergency department or chest pain unit. A literature search was performed in MEDLINE, EMBASE, Cochrane DSR, ACP Journal Club, DARE, CCTR, CMR, HTA, and NHSEED for studies from 2004 to 2010. They used the inclusion criteria:

Two reviewers conducted a hierarchical selection and assessment using a scale developed by the International Liaison Committee on Resuscitation. Out of a total 3,194 citations, 58 articles evaluating 37 novel biomarkers were included for final review. A total of 41 studies did not support the use of their respective biomarkers; 17 studies supported the use of 5 biomarkers, particularly when combined with cardiac-specific troponin: heart fatty acid-binding protein, ischemia-modified albumin, B-type natriuretic peptide, copeptin, and matrix metalloproteinase-9. The authors concluded that in patients presenting to the emergency department with chest pain or symptoms suggestive of cardiac ischemia, there is inadequate evidence to suggest the routine testing of novel biomarkers in isolation. Moreover, they stated that several novel biomarkers have the potential to improve the sensitivity of diagnosing ACS when combined with cardiac-specific troponin.

Eindhoven et al (2012) stated that BNP and NT-proBNP are well-established markers for heart failure in the general population. However, the value of BNP as a diagnostic and prognostic marker for patients with structural congenital heart disease (CHD) is still unclear. These investigators evaluated the clinical utility of BNP in patients with CHD. They executed a PubMed literature search and included 49 articles that focused on complex congenital heart defects such as tetralogy of Fallot, systemic right ventricle, and uni-ventricular hearts. Data on BNP measurements and cardiac function parameters were extracted. In all patients after correction for tetralogy of Fallot, BNP levels were elevated and correlated significantly with right ventricular end-diastolic dimensions and severity of pulmonary valve regurgitation. Patients with a systemic right ventricle had elevated BNP levels, and positive correlations between BNP and right ventricular function were seen. In patients with a uni-ventricular heart, elevated BNP levels were observed before completion of the Fontan circulation or when patients were symptomatic; a clear association between BNP and NYHA functional class was demonstrated. The authors concluded that this review showed an overall increase in BNP values in complex CHD, although differences between types of congenital heart anomaly are present. As BNP values differ widely, conclusions for individual patients should be drawn with

caution. They stated that further investigation with sequential BNP measurement in a large, prospective study is warranted to elucidate the prognostic value of BNP assessment in patients with CHD.

Eindhoven et al (2014) determined the value of NT-proBNP in adults with ToF and established its relationship with echocardiography and exercise capacity. Electrocardiography, detailed 2D-echocardiography and NT-proBNP measurement were performed on the same day in 177 consecutive adults with ToF (mean age of 34.6 ± 11.8 years, 58 % male, 89 % NYHA I, 29.3 ± 8.5 years after surgical correction); 38 % of the patients also underwent a cardiopulmonary-exercise test. Median NT-proBNP was 16 [interquartile range (IQR) 6.7 to 33.6] pmol/L, and was elevated in 55 %. NT-proBNP correlated with right ventricular (RV) dilatation ($r = 0.271$, $p < 0.001$) and RV systolic dysfunction ($r = -0.195$, $p = 0.022$), but more strongly with left-ventricular (LV) systolic dysfunction ($r = -0.367$, $p < 0.001$), which was present in 69 patients (39 %). Moderate or severe pulmonary regurgitation was not associated with higher NT-proBNP. Tricuspid and pulmonary regurgitation peak velocities correlated with NT-proBNP ($r = 0.305$, $p < 0.001$ and $r = 0.186$, $p = 0.045$, respectively). Left ventricular twist was measured with speckle-tracking echocardiography in 71 patients. An abnormal LV twist (20 patients, 28 %) was associated with elevated NT-proBNP ($p = 0.030$). No relationship between NT-proBNP and exercise capacity was found. The authors concluded that NT-proBNP levels were elevated in more than 50 % of adults with corrected ToF, while they were in stable clinical condition. Higher NT-proBNP is most strongly associated with elevated pulmonary pressures, and with LV dysfunction rather than RV dysfunction. They stated that NT-proBNP has the potential to become routine examination in patients with ToF to monitor ventricular function and may be used for timely detection of clinical deterioration.

García-Berrocso et al (2013) measured the association of BNP and NT-proBNP with all-cause mortality after stroke, and evaluated the additional predictive value of BNP/NT-proBNP over clinical information. Suitable studies for meta-analysis were found by searching MEDLINE and EMBASE databases until October 26, 2012. Weighted mean differences measured effect size; meta-regression and publication bias were assessed. Individual participant data were used to estimate effects by logistic regression and to evaluate BNP/NT-proBNP additional predictive value by area under the receiver operating characteristic curves, and integrated discrimination improvement and categorical net re-classification improvement indexes. Literature-based meta-analysis included 3,498 stroke patients from 16 studies and revealed that BNP/NT-proBNP levels were 255.78 pg/ml (95 % CI: 105.10 to 406.47, $p = 0.001$) higher in patients who died; publication bias entailed the loss of this association. Individual participant data analysis comprised 2,258 stroke patients. After normalization of the data, patients in the highest quartile had doubled the risk of death after adjustment for clinical variables (NIH Stroke Scale score, age, sex) (odds ratio 2.30, 95 % CI: 1.32 to 4.01 for BNP; and odds ratio 2.63, 95 % CI: 1.75 to 3.94 for NT-proBNP). Only NT-proBNP showed a slight added value to clinical prognostic variables, increasing discrimination by 0.028 points (integrated discrimination improvement index; $p < 0.001$) and reclassifying 8.1 % of patients into correct risk mortality categories (net re-classification improvement index; $p = 0.003$). Neither etiology nor time from onset to death affected the association of BNP/NT-proBNP with mortality. The authors concluded that BNPs are associated with post-stroke mortality independent of NIH Stroke Scale score, age, and sex. However, their translation to clinical practice seems difficult because BNP/NT-proBNP add only minor predictive value to clinical information. Thus, although this association was statistically significant, these biomarkers did not lead to better prediction of death than clinical information alone.

Hijazi et al (2013) assessed the prognostic value of NT-proBNP in patients with atrial fibrillation (AF) enrolled in the ARISTOTLE (Apixaban for the Prevention of Stroke in Subjects With Atrial Fibrillation) trial, and the treatment effect of apixaban according to NT-proBNP levels. In the ARISTOTLE trial, 18,201 patients with AF were randomized to apixaban or warfarin. Plasma samples at randomization were available from 14,892 patients. The association between NT-proBNP concentrations and clinical outcomes was evaluated using Cox proportional hazard models, after adjusting for established cardiovascular risk factors. Quartiles of NT-proBNP were: Q1, less than or equal to 363 ng/L; Q2, 364 to 713 ng/L; Q3, 714 to 1,250 ng/L; and Q4, greater than 1,250 ng/L. During 1.9 years, the annual rates of stroke or systemic embolism ranged from 0.74 % in the bottom NT-proBNP quartile to 2.21 % in the top quartile, an adjusted HR of 2.35 (95 % CI: 1.62 to 3.40; $p < 0.0001$). Annual rates of cardiac death ranged from 0.86 % in Q1 to 4.14 % in Q4, with an adjusted HR of 2.50 (95 % CI: 1.81 to 3.45; $p < 0.0001$). Adding

NT-proBNP levels to the CHA2DS2VASc score improved C-statistics from 0.62 to 0.65 ($p = 0.0009$) for stroke or systemic embolism and from 0.59 to 0.69 for cardiac death ($p < 0.0001$). Apixaban reduced stroke, mortality, and bleeding regardless of the NT-proBNP level. The authors concluded that NT-proBNP levels are often elevated in AF and independently associated with an increased risk of stroke and mortality. They stated that NT-proBNP improved risk stratification beyond the CHA2DS2VASc score and might be a novel tool for improved stroke prediction in AF. The effectiveness of apixaban compared with warfarin is independent of the NT-proBNP level.

In a prospective study, Rodriguez-Yanez et al (2013) examined if pro-BNP levels in the acute phase of stroke predict the development of AF in patients with cryptogenic stroke. Demographic data, medical history, and stroke characteristics were assessed at admission. A blood sample was obtained within the first 24 hours from stroke onset to determine pro-BNP levels. Patients were followed by a neurologist at 3 and 6 months and later by a primary care physician for 2 years to evaluate the development of AF. A total of 1,050 patients with ischemic stroke were evaluated -- 372 patients (35 %) had cryptogenic stroke. A total of 108 patients were excluded from the study, so 264 patients were valid for the analysis. Atrial fibrillation was detected in 15 patients (5.6 %) during the follow-up. Patients who developed AF were older, had hypertension more frequently, and showed higher levels of pro-BNP. In the logistic regression model, these researchers found that pro-BNP greater than or equal to 360 pg/ml was the only variable independently associated with the risk of developing AF (OR 5.70, 95 % CI: 1.11 to 29.29, $p = 0.037$). The authors concluded that pro-BNP greater than or equal to 360 pg/ml increases by 5-fold the possibility of detecting AF during follow-up in patients with cryptogenic stroke. They stated that pro-BNP levels of greater than or equal to 360 pg/ml determine during the acute phase of cryptogenic stroke may be useful to identify patients at risk of AF during the 2 years after the stroke. The drawbacks of this study included

Roldan et al (2014) stated that oral anti-coagulation is highly effective in reducing stroke and mortality in AF. Several risk stratification schemes have been developed using clinical characteristics. Elevated levels of NT-proBNP are important markers of increased mortality and morbidity in CHF and general community population. These investigators evaluated the predictive value of NT-proBNP levels in an unselected real-world cohort of anti-coagulated patients with AF. These researchers studied 1,172 patients (49 % male; median age of 76 years) with permanent AF who were well-stabilized on oral anti-coagulation (international normalized ratio, 2.0 to 3.0). Plasma NT-proBNP levels were quantified at baseline. These researchers recorded thrombotic and vascular events, mortality, and major bleeding. The best cut-off points were assessed by receiver-operating characteristic curves. Median levels (interquartile range) of NT-proBNP were 610 (318 to 1,037) pg/ml. Median follow-up was 1,007 (806 to 1,279) days. On multi-variate analysis, high NT-proBNP was significantly associated with the risk of stroke (HR, 2.71; $p = 0.001$) and composite vascular events (acute coronary syndrome or acute heart failure; HR, 1.85; $p = 0.016$), as well as a significant association with mortality (adjusted HR, 1.66; $p = 0.006$). No association with bleeding was found ($p = 0.637$). The integrated discrimination improvement (IDI) analysis demonstrated that NT-proBNP improved the CHF, Hypertension, Age greater than or equal to 75 (doubled), Diabetes mellitus, Stroke (doubled)-Vascular disease and Sex category (female); CHA2DS2-VASc score for predicting embolic events (relative IDI, 2.8 %; $p = 0.001$) and all-cause death (relative IDI, 1.8 %; $p = 0.001$). The authors concluded that in real-world cohort of anti-coagulated patients with AF, NT-proBNP provided complementary prognostic information to an established clinical risk score (CHA2DS2-VASc) for the prediction of stroke/systemic embolism. Moreover, they stated that NT-proBNP was also predictive of all-cause mortality, suggesting that this biomarker may potentially be used to refine clinical risk stratification in anti-coagulated patients with AF.

Balion et al (2014) stated that BNP/NT-proBNP measurement has not gained widespread use for the management of patients with HF despite several RCTs. These investigators performed a systematic review addressing the question of whether patients with HF benefit from BNP-assisted therapy or intensified therapy compared with usual care. Relevant RCTs were selected by searching Medline, Embase, AMED, Cochrane Central Register of Controlled Trials, Cochrane Database of Systematic Reviews, and CINAHL for English-language articles published from 1980 to 2012. Selected studies required patients to be treated for chronic HF with medical therapy based on BNP/NT-proBNP or usual care. There were no restrictions except that BNP/NT-proBNP measurement had to be done by an FDA-approved method. A total of 9 RCTs were identified with 2,104 patients with study duration that

ranged from 3 to 18 months. Overall, there was a wide variation in study design and how parameters were reported including patient selection, baseline characteristics, therapy goals, BNP/NT-proBNP cut-point, and outcome types. Meta-analysis was not appropriate given this study heterogeneity. The strength of evidence for the outcome of mortality, reported in 7 studies, was found to be low due to inconsistency and imprecision. The authors concluded that the findings of this systematic review showed that the evidence is of low quality and insufficient to support the use of BNP/NT-proBNP to guide HF therapy. They stated that further trials with improved design are needed.

Diagnosis of Cardio-Embolic Stroke

Yang et al (2014) performed a systematic review and meta-analysis to evaluate the value of BNP in differentiating cardio-embolic (CE) stroke from other subtypes of ischemic stroke. These investigators searched the EMBASE, MEDLINE, and Cochrane databases and reference lists of relevant articles published in April 2013. They selected original studies reporting the performance of BNP or NT-proBNP in diagnosing CE stroke and summarized test performance characteristics using forest plots, hierarchical summary receiver operating characteristic curves, and bivariate random-effect models. Data from 2,958 patients with ischemic stroke were retrieved from 16 studies. Of these, 1,024 (34.6 %) patients had a final diagnosis of CE stroke. Overall, the mean diagnostic OR (DOR) of BNP for CE stroke was 15.8 (95 % CI: 9.92 to 25.20). Even after adjustment for multiple clinical predictors, serum natriuretic peptide levels showed a strong association with CE stroke (pooled adjusted DOR, 12.7; 95 % CI: 7.32 to 22.0). The sensitivity and specificity of BNP for CE stroke were 0.78 (95 % CI: 0.71 to 0.87) and 0.83 (95 % CI: 0.77 to 0.87), respectively. A single BNP-negative result may be sufficient to exclude a diagnosis of CE stroke in low-prevalence (less than 20 %) settings. Subgroup analysis showed that NT-proBNP had a slightly higher specificity (0.87; 95 % CI: 0.77 to 0.93) and better capability for exclusion diagnosis. There was a lack of homogeneity in the timing of measurement and BNP assay method. The authors concluded that BNP has reasonable accuracy in the diagnosis of CE stroke and may be a useful marker for the early detection in patients who may benefit from preventive anti-coagulation therapy.

An UpToDate review on "Clinical diagnosis of stroke subtypes" (Caplan, 2015a) does not mention the use of measurement of plasma brain natriuretic peptide as a diagnostic tool. Furthermore, an UpToDate review on "Overview of the evaluation of stroke" (Caplan, 2015b) states that "Other potential indicators that may predict which patients have or are likely to develop atrial fibrillation include determination of the left atrial appendage ejection fraction by transesophageal echocardiography and demonstration of an atrial fibrillation phenotype on left atrial appendage pulse wave Doppler even when the surface electrocardiogram (ECG) shows normal sinus rhythm. In addition, measurements of B-type (brain) natriuretic protein (BNP) and the N-terminal fragment of BNP indicate that these values are elevated in patients with atrial fibrillation who have cardiogenic embolism, even in those with normal ventricular function, when compared with patients who have non-cardiogenic stroke. These measurements in patients with normal cardiac ventricular function might identify those who have intermittent atrial fibrillation or are prone to develop it. Further research is needed to confirm whether these indicators are clinically useful".

Llombart et al (2015) noted that increased blood levels of BNP/NT-proBNP have been repeatedly associated with cardio-embolic stroke. These investigators evaluated their clinical value as pathogenic biomarkers for stroke through a literature systematic review and individual participants' data meta-analysis. They searched publications in PubMed database until November 2013 that compared BNP and NT-proBNP circulating levels among stroke causes. Standardized individual participants' data were collected to estimate predictive values of BNP/NT-proBNP for cardio-embolic stroke. Dichotomized BNP/NT-proBNP levels were included in logistic regression models together with clinical variables to assess the sensitivity and specificity to identify cardio-embolic strokes and the additional value of biomarkers using area under the curve and integrated discrimination improvement index. From 23 selected articles, these researchers collected information of 2,834 patients with a defined cause; BNP/NT-proBNP levels were significantly elevated in cardio-embolic stroke until 72 hours from symptoms onset. Predictive models showed a sensitivity greater than 90 % and specificity greater than 80 % when BNP/NT-proBNP were added considering the lowest and the highest quartile, respectively. Both peptides also increased significantly the area under the curve and integrated discrimination improvement index compared with clinical models. Sensitivity, specificity, and precision of the models were validated in 197 patients with initially undetermined stroke with

final pathogenic diagnosis after ancillary follow-up. The authors concluded that natriuretic peptides are strongly increased in cardio-embolic strokes. Moreover, they stated that future multi-center prospective studies comparing BNP and NT-proBNP might aid in finding the optimal biomarker, the best time-point, and the optimal cut-off points for cardio-embolic stroke identification.

Diagnosis of Patent Ductus Arteriosus

Farombi-Oghuvbu et al (2008) noted that BNP is a marker for ventricular dysfunction secreted as a pre-prohormone, proBNP, and cleaved into BNP and a biologically inactive fragment, NT-proBNP. Little is known about the clinical usefulness of NT-proBNP in preterm infants. These researchers evaluated the usefulness of plasma NT-proBNP in diagnosing hemodynamically significant patent ductus arteriosus (hsPDA) in neonates and examined some factors that might affect this. Infants born at less than 34 weeks' gestational age (GA) and less than 2 kg birth weight (BW) were prospectively enrolled within 6 to 12 hours of birth. Plasma NT-proBNP levels were measured on days 1, 3, 5 and 10 with simultaneous echocardiography done to detect hsPDA and assess ventricular function. Significant PDA was diagnosed by large ductal flow with left to right shunt on color Doppler, measuring greater than 1.6 mm on 2-dimensional echocardiography, along with clinical features of PDA. A total of 49 infants were analyzed. Median GA was 30 weeks (range of 24 to 33) and median BW 1,220 g (range of 550 to 1,950). Eighteen infants with hsPDA had higher day 3 plasma NT-proBNP values (median of 32,907 pg/ml; range of 11,396 to 127,155) ($p < 0.001$) than controls (median of 3,147 pg/ml; range of 521 to 10,343). Infants who developed sepsis had higher day 10 plasma NT-proBNP levels.

Area under receiver operator characteristic curve for detection of hsPDA, by day 3 NT-proBNP value, was significant 0.978 (95 % CI: 0.930 to 1.026). NT-proBNP was predictive of hsPDA (sensitivity 100 %; specificity 95 %) at a cut-off value of 11,395 pg/ml. The authors concluded that plasma NT-proBNP level on day 3 is a good marker for hsPDA in preterm infants; serial measurements of NT-proBNP may be useful in assessing the clinical course of PDA.

Kulkarni et al (2015) stated that echocardiogram is the gold standard for the diagnosis of hsPDA in preterm neonates. A simple blood assay BNP or NT-proBNP may be useful in the diagnosis and management of hsPDA. These researchers determined the diagnostic accuracy of BNP and NT-proBNP for hsPDA in preterm neonates and explored heterogeneity by analyzing subgroups. The systematic review was performed as recommended by the Cochrane Diagnostic Test Accuracy Working Group. Electronic databases, conference abstracts, and cross-references were searched. These investigators included studies that evaluated BNP or NT-proBNP (index test) in preterm neonates with suspected hsPDA (participants) in comparison with echocardiogram (reference standard). A bivariate random effects model was used for meta-analysis, and summary receiver operating characteristic curves were generated. A total of 10 BNP and 11 NT-proBNP studies were included. Studies varied by methodological quality, type of commercial assay, thresholds, age at testing, gestational age, and whether the assay was used to initiate medical or surgical therapy. Sensitivity and specificity for BNP at summary point were 88 % and 92 %, respectively, and for NT-proBNP they were 90 % and 84 %, respectively. The authors concluded that studies evaluating the diagnostic accuracy of BNP and NT-proBNP for hsPDA varied widely by assay characteristics (assay kit and threshold) and patient characteristics (gestational and chronological age); therefore, generalizability between centers is not possible. They recommended that BNP or NT-proBNP assays be locally validated for specific patient population and outcomes, to initiate therapy or follow response to therapy.

Furthermore, an UpToDate reviews on "Pathophysiology, clinical manifestations, and diagnosis of patent ductus arteriosus in premature infants" (Phillips, 2015) states that "Biomarkers, especially B-type natriuretic peptide (BNP) or the inactive N-terminal pro-BNP, which has a longer half-life, have been proposed as useful in the diagnosis and management of PDA. However, the sensitivity and specificity of these tests vary in different populations and sites and further study is needed to identify their role in the diagnosis of PDA".

Diagnosis of Kawasaki Disease

In a systematic review and meta-analysis, Lin and colleagues (2015) examined the diagnostic value of serum BBNP in acute Kawasaki disease (KD). A systematic literature search strategy was designed and carried out using Medline, Embase and the Cochrane Library from inception to December 2013. These investigators also performed manual screening of the bibliographies of primary studies and review articles, and contacted

authors for additional data. They included all BNP and NT-proBNP assay studies that compared pediatric patients with KD to patients with febrile illness unrelated to KD. These researchers excluded case reports, case series, review articles, editorials, congress abstracts, clinical guidelines and all studies that compared healthy controls. The performance characteristics of BNP were summarized using forest plots, hierarchical summary receiver operating characteristic (ROC) curves and bivariate random effects models. The authors found 6 eligible studies including 279 cases of patients with KD and 203 febrile controls; 6 studies examined NT-proBNP and 1 examined BNP. In general, NT-proBNP is a specific and moderately sensitive test for identifying KD. The pooled sensitivity was 0.89 (95 % CI: 0.78 to 0.95) and the pooled specificity was 0.72 (95 % CI: 0.58 to 0.82). The area under the summary ROC curve was 0.87 (95 % CI: 0.83 to 0.89). The positive likelihood ratio (LR+ 3.20, 95 % CI: 2.10 to 4.80) was sufficiently high to be qualified as a rule-in diagnostic tool in the context of high pre-test probability and compatible clinical symptoms. A high degree of heterogeneity was found using the Cochran Q statistic. The authors concluded that current evidence suggested that NT-proBNP may be used as a diagnostic tool for KD; NT-proBNP had high diagnostic value for identifying KD in patients with protracted undifferentiated febrile illness.

Moreover, they stated that prospective large cohort studies are needed to help determine best cut-off values and further clarify the role of NT-proBNP in the diagnosis of KD.

In a systematic review and meta-analysis, Wen et al (2021) examined the diagnostic accuracy of circulating NT-proBNP for Kawasaki disease (KD). These investigators searched the PubMed, Web of Science and EMBASE databases to identify the eligible studies investigating the diagnostic accuracy of NT-proBNP for KD. The revised tool for the quality assessment of diagnostic accuracy studies (QUADAS-2) was used to evaluate the eligible studies' quality. A meta-analysis was carried out with the bi-variate model and summary ROC (sROC) curve. These researchers also performed subgroup, publication bias and sensitivity analyses. They included 12 studies with 2,173 KDs and 1,909 control.

The pooled sensitivity and specificity of eligible studies were 0.80 (95 % CI: 0.72 to 0.86) and 0.81 (95 % CI: 0.73 to 0.88), respectively. The area under sROC curve was 0.88 (95 % CI: 0.84 to 0.90). Patient selection bias and partial verification bias were the major design limitations of the eligible studies. Sensitivity analysis revealed that the results of this meta-analysis were robust. Subgroup analysis revealed that study design, NT-proBNP assay and subjects' body temperature were not the source of heterogeneity across all eligible studies. No publication bias was observed. The authors concluded that NT-proBNP had moderate diagnostic accuracy for KD; however, it could not be used for ruling in or ruling out KD when used alone. Moreover, these researchers stated that further well-designed studies are needed to examine the diagnostic accuracy of NT-proBNP for KD.

The authors stated that this systematic review and meta-analysis had 3 limitations. The 1st limitation was that all eligible studies were from Asia and North America; thus, investigators should be cautious about extending this study's results to areas other than Asia and North American. The 2nd limitation was the great heterogeneity across all eligible studies. Although these researchers had examined the source of heterogeneity with subgroup analysis and sensitivity analysis, it remained unknown. The 3rd limitation was that some of the eligible studies were not reported following the Standards for Reporting of Diagnostic Accuracy Studies (STARD) guideline; thus, these investigators could not comprehensively evaluate the quality of the eligible studies.

Identification of Individuals at Risk of Developing Abnormal Brain Aging

In across-sectional study, Sabayan et al (2015) examined the independent association of serum NT-proBNP with structural and functional features of abnormal brain aging in older individuals. This study was based on the Age, Gene/Environment Susceptibility (AGES)-Reykjavik Study, and these investigators included 4,029 older community-dwelling individuals (born 1907 to 1935) with a measured serum level of NT-proBNP. Outcomes included parenchymal brain volumes estimated from brain magnetic resonance imaging (MRI), cognitive function measured by tests of memory, processing speed, and executive functioning, and presence of depressive symptoms measured using the Geriatric Depression Scale. In a sub-study, cardiac output of 857 participants was assessed using cardiac MRI. In multi-variate analyses, adjusted for socio-demographic and cardiovascular factors, higher levels of NT-proBNP were independently associated with lower total ($p < 0.001$), gray matter ($p < 0.001$), and white matter ($p = 0.001$) brain volumes. Likewise, in multivariate analyses, higher levels of NT-proBNP were associated with worse scores in memory ($p = 0.005$), processing speed ($p = 0.001$), executive functioning ($p < 0.001$), and more depressive symptoms ($p = 0.002$). In the sub-study, the associations of higher NT-

proBNP with lower brain parenchymal volumes, impaired executive function and processing speed, and higher depressive symptoms were independent of the level of cardiac output. The authors concluded that higher serum levels of NT-proBNP, independent of cardiovascular risk factors and a measure of cardiac function, are linked with alterations in brain structure and function. Moreover, they stated that the roles of natriuretic peptides in the process of brain aging need to be further elucidated. They noted that further research is needed to elucidate mechanisms underlying this association and to clarify whether measurement of NT-proBNP can be a tool to identify older individuals at high risk of developing abnormal brain aging.

Prediction of the Occurrence of Atrial Fibrillation after Thoracic Surgery

In a systematic review and meta-analysis, Simmers et al (2015) examined if elevated pre-operative BNP measurements are an independent predictor of AF in patients having thoracic surgery. Embase, Ovid Health Star, Ovid Medline, Cochrane Central Register of Controlled Trials, Cochrane Database of Systematic Reviews, and ProQuest Dissertations and Theses A&I databases were searched for all studies of non-cardiac thoracic surgery patients in whom a pre-operative NP was measured up to 1 month before surgery, and that measured the incidence of post-operative AF. Studies were included regardless of their language, sample size, publication status, or study design. Study quality was evaluated using the Newcastle Ottawa Scale. The combined incidence of post-operative AF was 14.5 % (n = 108/742), and the NP thresholds used to predict AF varied among studies. An elevated pre-operative NP measurement was associated with an OR of 3.13 (95 % CI: 1.38 to 7.12; I² = 87 %) for post-operative AF, with the sensitivity analysis reporting an OR of 9.51 (95 % CI: 4.66 to 19.40; I² = 0). The authors concluded that patients with an elevated pre-operative NP measurement were at an increased risk of post-operative AF. They stated that there may be value in incorporating NP measurement into existing AF risk prediction models.

Prediction of Outcome in Congenital Diaphragmatic Hernia

Snoek et al (2016) stated that biomarkers may be helpful in prediction of outcomes of infants with congenital diaphragmatic hernia (CDH). The predictive value of high-sensitivity troponin T and NT-proBNP was investigated in 128 infants with CDH. After correction for multiple testing, these biomarkers did not predict severe pulmonary hypertension, death, need of extra-corporeal membrane oxygenation, or broncho-pulmonary dysplasia.

Risk Stratification of Individuals with Aortic Stenosis

Lindman et al (2015) examined if multiple biomarkers of cardiovascular stress are associated with mortality in patients with aortic stenosis (AS) undergoing aortic valve replacement (AVR) independent of clinical factors. From a prospective registry of patients with AS, a total of 345 participants who were referred for and treated with AVR (trans-catheter [n = 183] or surgical [n = 162]) were included. A total of 8 biomarkers were measured on blood samples obtained prior to AVR: growth differentiation factor 15 (GDF15), soluble ST2 (sST2), NT-proBNP, galectin-3, high-sensitivity cardiac troponin T, myeloperoxidase, high-sensitivity C reactive protein and monocyte chemoattractant protein-1. Biomarkers were evaluated based on median value (high versus low) in a Cox proportional hazards model for all-cause mortality and a parsimonious group of biomarkers selected.

Mean follow-up was 1.9 ± 1.2 years; 91 patients died. Three biomarkers (GDF15, sST2 and NT-proBNP) were retained in the model. One-year mortality was 5 %, 12 %, 18 % and 33 % for patients with 0 (n = 79), 1 (n = 96), 2 (n = 87) and 3 (n = 83) biomarkers elevated, respectively (p < 0.001). After adjustment for the Society of Thoracic Surgeons (STS) risk score, a greater number of elevated biomarkers was associated with increased mortality (referent: 0 elevated): 1 elevated (HR 1.47, 95 % CI: 0.60 to 3.63, p = 0.40), 2 elevated (HR 2.89, 95 % CI: 1.24 to 6.74, p = 0.014) and 3 elevated (HR 4.59, 95 % CI: 1.97 to 10.71, p < 0.001). Among patients at intermediate or high surgical risk (STS score greater than or equal to 4), 1-year and 2-year mortality rates were 34 % and 43 % for patients with 3 biomarkers elevated versus 4 % and 4 % for patients with 0 biomarkers elevated. When added to the STS score, the number of biomarkers elevated provided a category-free net re-classification improvement of 64 % at 1 year (p < 0.001). The association between a greater number of elevated biomarkers and increased mortality after valve replacement was similar in the trans-catheter and surgical AVR populations. The authors concluded that the findings of this study demonstrated the potential utility of multiple biomarkers to aid in risk stratification of patients with AS. Moreover, they

stated that further studies are needed to evaluate their utility in clinical decision-making in specific AS populations.

Biomarker for Cerebral Small Vessel Disease/Vascular Brain Damage in Hypertension

Vilar-Bergua and colleagues (2016) studied the association of NNT-proBNP with several brain MRI markers of brain vascular disease in a sample of subjects free of stroke and dementia. Plasma level of NT-proBNP was determined by means of a sandwich immunoassay method in a cohort study comprising 278 hypertensive patients. The presence of silent brain infarcts (SBIs), brain micro-bleeds, enlarged peri-vascular spaces, and white matter hyper-intensity volumes was assessed by brain MRI. These researchers performed uni-variate and multi-variate analyses to examine if NT-proBNP was independently associated with these imaging markers, individually or combined. Median age was 63 years, and 41.4 % were women; NT-proBNP remained independently associated with silent brain infarcts (OR per 1-SD increase in NT-proBNP 2.11, 95 % CI: 1.44 to 3.10), brain micro-bleeds (OR 1.79, 95 % CI: 1.15 to 2.78), basal ganglia enlarged peri-vascular spaces (OR 1.55, 95 % CI: 1.12 to 2.15), and white matter hyper-intensity volumes (β 1.60, 95 % CI: 0.47 to 2.74), even after controlling for vascular risk factors, cardiovascular risk, AF, previous heart disease, duration of hypertension, and preventive treatments. A score combining several imaging markers was also related to NT-proBNP levels (common OR per 1-SD increase 1.74, 95 % CI: 1.21 to 2.50). The authors concluded that these findings suggested that NT-proBNP plasma levels relate to several subclinical MRI markers of cerebral small vessel disease (CSVD) and to their burden. They stated that these results might be useful to identify individuals more likely to present CSVD lesions and to advance the prevention of stroke and dementia; more research is needed to enable the use of NT-proBNP as a potential biomarker to detect CSVD in the brain.

The authors stated that this study had several drawbacks. First, although the Investigating Silent Strokes in Hypertensives (ISSYS) cohort was composed of randomly selected hypertensives from the community, in this study, men with SBIs were over-represented, and this should be taken into account in the generalization of these findings. Also, these researchers did not identify a cut-off of NT-proBNP that better predicted the presence of each of the brain lesions of interest or the burden of these lesions in this cohort. Thus, despite the associations between NT-proBNP and the MRI markers studied here, more research is needed to enable its use as a potential biomarker to detect CSVD in the brain. Furthermore, although these analyses were adjusted for the presence of prior or actual coronary artery disease or HF, a measure of specific cardiac function was not obtained in this cohort. Finally, the cross-sectional design of this study prevented these investigators from establishing causal relationships.

Biomarker of Clinical Improvement of Heart Failure After Transluminal Septal Myocardial Ablation for Drug-Refractory Hypertrophic Obstructive Cardiomyopathy

Akita and colleagues (2018) examined if repeated BNP measurements after percutaneous transluminal septal myocardial ablation (PTSMA) provide prognostic information regarding the response to PTSMA in patients with drug-refractory hypertrophic obstructive cardiomyopathy (HOCM). These researchers measured the plasma BNP levels serially before and after PTSMA, and evaluated the relationship between the changes in plasma BNP levels and clinical improvement in 47 patients. Participants were assigned to 2 groups based on the reduction in the NYHA class greater than or equal to 1 (good responder) or less than 1 (poor responder) before and after PTSMA. The Kansas City Cardiomyopathy Questionnaire (KCCQ) was used to measure health status. Plasma BNP levels gradually decreased after PTSMA, although the levels plateaued 3 months until 12 months after PTSMA. Although the plasma BNP levels and resting left ventricular outflow tract peak pressure gradient before PTSMA were comparable between the groups, the ratio of the BNP levels before and after PTSMA in the good responder group was significantly lower than that in the poor responder group (0.43 (range of 0.24 to 0.68) versus 0.78 (range of 0.62-0.93), $p = 0.002$). The KCCQ score changes in the good responder group were significantly higher than those in the poor responder group. The authors concluded that plasma BNP level ratio was associated with long-term clinical improvement of HF after PTSMA for drug-refractory HOCM. Moreover, they stated that a larger study is needed to elucidate the profound relationship between the ratio of BNP and HF symptoms before and after PTSMA.

They stated that these findings might provide a more careful and convenient follow-up method after PTSMA and act as a foundation for future large cohort studies focused on the relationship between the BNP ratio and septal reduction therapy.

The authors stated that this study had several drawbacks. First, the study population was of limited size ($n = 47$), although truly drug-refractory HOCM patients are relatively uncommon. However, serial measurements of the BNP levels over 12 months were performed, and the PTSMA technique was consistent among these patients. A previous report showed that a consistent PTSMA technique was important and was associated with superior outcomes with alcohol septal ablation. Therefore, this study used a uniform study platform, thus making it possible to collect reliable data and reducing the limitations of a limited sample size. Second, these findings may not be generalizable to the global HOCM population, as the patients were treated at a hospital with extensive experience in treating HOCM. Third, this study showed selection bias, as these investigators selected patients with HOCM who were suitable for PTSMA.

Biomarker for Subclinical Brain Damage

In a prospective, population-based, cohort study, Zonneveld and associates (2017) examined the association between NT-proBNP and markers of subclinical brain damage on MRI in community-dwelling middle-aged and elderly subjects without dementia and without a clinical diagnosis of heart disease. Serum levels of NT-proBNP were measured in 2,397 participants without dementia or stroke (mean age of 56.6 years; age range of 45.7 to 87.3 years) and without clinical diagnosis of heart disease who were drawn from the population-based Rotterdam Study. All participants were examined with a 1.5-T MR imager.

Multi-variable linear and logistic regression analyses were used to examine the association between NT-proBNP level and MRI markers of subclinical brain damage, including volumetric, focal, and microstructural markers. A higher NT-proBNP level was associated with smaller total brain volume (mean difference [MD] in z score per standard deviation increase in NT-proBNP level, -0.021 ; 95 % CI: -0.034 to -0.007 ; $p = 0.003$) and was predominantly driven by gray matter volume (MD in z score per standard deviation increase in NT-proBNP level, -0.037 ; 95 % CI: -0.057 to -0.017 ; $p < 0.001$). Higher NT-proBNP level was associated with larger white matter lesion volume (MD in z score per standard deviation increase in NT-proBNP level, 0.090 ; 95 % CI: 0.051 to 0.129 ; $p < 0.001$), with lower fractional anisotropy (MD in z score per standard deviation increase in NT-proBNP level, -0.048 ; 95 % CI: -0.088 to -0.008 ; $p = 0.019$) and higher mean diffusivity (MD in z score per standard deviation increase in NT-proBNP level, 0.054 ; 95 % CI: 0.018 to 0.091 ; $p = 0.004$) of normal-appearing white matter. The authors found that in community-dwelling middle-aged and elderly persons, subclinical cardiac dysfunction as reflected by serum NT-proBNP levels was associated with global and microstructural MR imaging markers of subclinical brain damage. They stated that these findings suggested that the heart and brain are intimately linked, even in presumably healthy individuals. Moreover, they stated that further research is needed to elucidate the causal relationship between cardiac dysfunction and subclinical brain disease and to explore the prevailing pathway among several hypotheses. Additionally, it may be interesting to examine if NT-proBNP could be used as a clinically relevant marker for subclinical brain damage.

The authors stated that this study had 2 main drawbacks. First, this trial was a cross-sectional study; thus, these researchers could not draw conclusions regarding causality or the direction of the associations. However, from a biologic perspective, and on the basis of animal studies, it is more likely that cardiac dysfunction affects brain changes than vice versa. Second, this study consisted largely of white participants, and this may have limited extrapolation of these findings to patients of other ethnicities.

Prognostic Biomarker in Acute Coronary Syndromes

Eggers and colleagues (2017) noted that cardiac troponin (cTn) plays an essential role for assessment of outcome in acute coronary syndrome (ACS). However, the prognostic value of cTn is not absolute. In this mini-review, these investigators summarized the evidence on the utility of established biomarkers of LV dysfunction, hemodynamic stress, inflammation, and renal dysfunction for risk prediction beyond cTn in ACS. Only few biomarkers consistently demonstrated additive prognostic value to cTn levels. The B-type natriuretic peptides (NPs) and growth-differentiation factor-15 (GDF-15) are most promising in this regard. However, there are uncertainties regarding the role of these biomarkers for guidance of treatment decisions, and their prognostic increment to cTn levels measured with high-sensitivity assays is largely unknown. The authors concluded that the NPs and GDF-15 provide the strongest prognostic increment to cTn levels in ACS.; however, the role of these biomarkers for clinical decision-making in contemporary settings has still to be defined.

Furthermore, an UpToDate review on "Overview of the acute management of non-ST elevation acute coronary syndromes" (Aroesty et al, 2018) does not mention B-type natriuretic peptide as a biomarker.

Detection of Early Cardiac Dysfunction in Individuals with Chronic Fatigue Syndrome

In a case control study, Tomas and colleagues (2017) examined levels of the BNP and how they may be associated with the cardiac abnormalities in chronic fatigue syndrome (CFS). Cardiac MRIs were performed using 3T Philips Intera Achieva scanner (Best, Netherlands) in CFS participants and sedentary controls matched group-wise for age and sex; BNP was also measured by using an enzyme immunoassay in plasma from 42 patients with CFS and 10 controls. BNP levels were significantly higher in the CFS cohort compared with the matched controls ($p = 0.013$). When these researchers compared cardiac volumes (end-diastolic and end-systolic) between those with high BNP levels (BNP greater than 400 pg/ml) and low BNP (less than 400 pg/ml), there were significantly lower cardiac volumes in those with the higher BNP levels in both end-systolic and end-diastolic volumes ($p = 0.05$). There were no relationships between fatigue severity, length of disease and BNP levels ($p = 0.2$) suggesting that these findings were unlikely to be related to deconditioning. The authors concluded that the findings of this study confirmed an association between reduced cardiac volumes and BNP in CFS. They stated that lack of relationship between length of disease and BNP levels suggested that findings were not secondary to deconditioning. Moreover, they stated that further studies are needed to examine the utility of BNP to act as a stratification paradigm in CFS that directs targeted treatments.

Diagnosis, prognostic Evaluation and Screening for Hypertrophic Cardiomyopathy

To test dual blood biomarkers compared with ECG for hypertrophic cardiomyopathy (HC) screening, Blackshear and colleagues (2018) performed 3 analyses and cut-point assessments. First, these researchers measured platelet function analyzer (PFA)-100 ($n = 99$) and normalized BBNP or NT-proBNP (BNP/upper limit of normal [ULN], $n = 92$) in 64 patients with HC and 29 normal controls (NCs). Second, from the regression equation between PFA and gradient ($r = 0.77$), these investigators derived estimated PFA in a population of 189 patients with functional class I HC in whom measured BNP/ULN and ECG were available, and calculated single and dual biomarker sensitivity and specificity compared with ECG. Finally, these researchers compared BNP/ULN in class I patients based on mutation and familial history status. In 42 patients with obstructive HC versus NCs, there was a slight overlap of PFA and BNP/ULN, but for the product of PFA $\times$ BNP/ULN, there was near-complete separation of values. Among patients with class I obstructive HC, estimated PFA $\times$ BNP/ULN had a sensitivity of 93 % and a specificity of 100 %; in latent and non-obstructive HC, sensitivity dropped to 61 % and 72 %; for ECG in obstructive, latent, and non-obstructive HC, sensitivity was 71 %, 34%, and 67%. Functional class I patients with positive ($n = 28$) and negative ($n = 36$) sarcomere mutations and a positive ($n = 71$) or a negative ($n = 109$) family history had significant elevations of BNP/ULN versus NC, with no between-group differences. The authors concluded that PFA and BNP were highly associated with obstructive HC and could potentially be used for screening; BNP was not uniquely elevated in patients with familial versus non-familial or mutation-positive versus mutation-negative HC.

An UpToDate review on "Hypertrophic cardiomyopathy: Clinical manifestations, diagnosis, and evaluation" (Maron, 2018) states that "The range of values associated with plasma brain natriuretic peptide (BNP) and N-terminal pro-BNP concentration is quite broad and does not correlate well with HF symptoms in patients with HCM. As a result, we do not order this test as part of the diagnostic or prognostic evaluation of patients with suspected HCM".

However, guidelines on hypertrophic cardiomyopathy from the European Society of Cardiology (Elliott, et al., 2014) recommend measurement of BNP. "High levels of brain natriuretic peptide (BNP), N-terminal pro-brain natriuretic peptide (NT-proBNP) and high sensitivity cardiac troponin T (hs-cTnT) are associated with cardiovascular events, heart failure and death. Despite comparable values of ventricular wall thickness, plasma BNP values are three to five times as high in patients with cardiac amyloidosis as those in other causes of HCM."

Diagnosis of Preeclampsia

Ortner and colleagues (2019) stated that pilot studies applying point-of-care ultra-sound

(POCUS) in preeclampsia indicated the presence of pulmonary interstitial edema, cerebral edema, and cardiac dysfunction. Laboratory markers of oncotic pressure (albumin) and cardiac dysfunction (BNP) may be abnormal, but the clinical application remains unclear.

In a prospective, observational, cohort study, these investigators examined the prevalence of pulmonary interstitial syndrome (PIS), cardiac dysfunction, and increased optic nerve sheath diameter (ONSD) in late-onset preeclampsia with severe features. The primary objective was to determine the association between PIS or ONSD and maternal serum albumin level. The secondary objectives were to examine the association between cardiac dysfunction and PIS, ONSD, BNP, and serum albumin level and between POCUS-derived parameters and a suspicious or pathological cardiotocograph. A total of 95 women were enrolled in this trial. A POCUS examination of lungs, heart, and ONSD was performed; PIS was defined as a bilateral B-line pattern on lung US and diastolic dysfunction according to an algorithm of the American Society of Echocardiography; ONSD greater than 5.8 mm was interpreted as compatible with raised intra-cranial pressure (greater than 20 mm Hg).

Serum BNP and albumin levels were also measured. PIS, diastolic dysfunction, systolic dysfunction, and raised left ventricular end-diastolic pressure (LVEDP) were present in 23 (24 %), 31 (33 %), 9 (10 %), and 20 (25 %) women, respectively. ONSD was increased in 27 (28 %) women. Concerning the primary outcome, there was no association between albumin level and PIS ($p = 0.4$) or ONSD ($p = 0.63$). With respect to secondary outcomes, there was no association between albumin level and systolic dysfunction ($p = 0.21$) or raised LVEDP ($p = 0.44$). PIS was associated with diastolic dysfunction ($p = 0.02$) and raised LVEDP ($p = 0.009$; negative predictive value [NPV], 85 %). BNP level was associated with systolic ($p < 0.001$) and diastolic dysfunction ($p = 0.003$) and LVEDP ($p = 0.007$). No association was found between POCUS abnormalities and a suspicious/pathological cardiotocograph ($p = 0.07$). The authors concluded that PIS, diastolic dysfunction, and increased ONSD were common in preeclampsia with severe features.

Cardiac US abnormalities may be more useful than albumin levels in predicting PIS. The absence of PIS may exclude raised LVEDP. The further clinical relevance of PIS and raised ONSD remains to be established. BNP level was associated with cardiac US abnormalities. These researchers stated that although this study was not designed to directly influence clinical management, the findings suggested that POCUS may serve as a useful adjunct to clinical examination for the obstetric anesthesiologist managing these complex patients.

Furthermore, an UpToDate review on "Preeclampsia: Clinical features and diagnosis" (August and Sibai, 2019) does not mention measurement of serum BNP as a diagnostic tool.

Screening or Diagnosis of Pulmonary Hypertension Associated with Bronchopulmonary Dysplasia

Konig and associates (2016) stated that bronchopulmonary dysplasia (BPD) is often complicated by pulmonary hypertension (PH). In a prospective, observational cohort study, these investigators examined 3 biomarkers potentially suitable as screening markers for extremely pre-term infants at risk of BPD-associated PH. This trial was carried out in a tertiary neonatal intensive care unit (ICU). A total of 83 pre-term infants with BPD born less than 28-week gestation and still in-patients at 36-week corrected age received an echocardiogram (echo) and blood tests of BNP, troponin I, and YKL-40. Infants were analyzed according to echocardiographic evidence of tricuspid regurgitation (TR); 30 infants had evidence of TR on echo at 36-week corrected age. Infants with or without TR had similar baseline demographics: mean $\pm$ SD GA of 261 ± 12 versus 261 ± 11 weeks and birth-weight of 830 ± 206 versus 815 ± 187 g, respectively.

There was no difference in duration of respiratory support. The right ventricular systolic pressure of infants with evidence of TR was 40 ± 16 mmHg; BNP was the only biomarker that proved to be significantly higher in infants with evidence of TR: median (IQR) serum level 54.5 (35 to 105) versus 41.5 (30 to 59) pg/ml, $p = 0.043$. Subgroup analysis of infants with severe BPD requiring discharge on home oxygen or BPD-related mortality revealed similar results. There was no difference between groups for troponin I and YKL-40. The authors concluded that increased serum levels of BNP were associated with evidence of TR at 36-week corrected GA in extremely pre-term infants, suggesting a potential role as a screening biomarker for BPD-associated PH.

Avitabile and colleagues (2019) noted that premature infants with severe BPD (sBPD) are at risk of PH. Serum BNP is used to predict disease severity in adult PH. Its diagnostic utility in sBPD-associated PH is unknown. These investigators examined the accuracy of BNP, against echo, to diagnose PH in infants born less than 32 weeks' gestation with sBPD. They carried out a retrospective cohort study of all infants with

sBPD with an echo and BNP within a 24-hour period, at greater than or equal to 36 weeks post-menstrual age. Pulmonary hypertension was defined as: right ventricular pressure greater than $\frac{1}{2}$ systemic blood pressure (BP) estimated from TR jet or PDA velocity, bi-directional or right-to-left-PDA, and/or flat/bowing ventricular septum at end-systole.

Receiver-operating characteristic (ROC) curves were constructed to test the diagnostic accuracy of BNP. Of 128 infants, 68 (53 %) had echo evidence of PH; BNP was higher among the infants with PH (median [IQR]: 127 pg/ml [39 to 290] versus 35 [20 to 76], $p < 0.001$). The area under the ROC curve for diagnosing PH using BNP was 0.74 (95 % CI: 0.66 to 0.83). At an optimal cut-point of 130 pg/ml, BNP correctly classified the presence or absence of PH in 70 % of the infants (specificity: 92 %, sensitivity: 50 %). The authors concluded that BNP, relative to concurrent echo, demonstrated moderate accuracy for diagnosing PH in this cohort of pre-term infants with sBPD. These researchers stated that BNP may help rule in PH in this population; but has low utility to rule out the disease.

In a retrospective, longitudinal, cohort study, Behere and co-workers (2019) examined the ability of routine neonatal screening at time of BPD diagnosis to predict the development of late PH. This trial ($n = 37$ premature infants with BPD) evaluated the utility of screening serum BNP and echo performed at the time of BPD diagnosis ("early PH") to predict "late PH" at the last follow-up. Screening evaluation demonstrated early PH in 9/37 patients. At an average follow-up interval of 52.7 ± 38.7 weeks, 4/9 had late PH; 1 patient without early PH had late PH. At initial screening, infants with late PH were significantly more likely to have demonstrated elevated BNP values ($p = 0.003$), and echocardiographic evidence of right atrial dilatation ($p = 0.01$), right ventricular hypertrophy ($p = 0.01$), lower right ventricular area change percentage ($p = 0.03$), and larger main pulmonary artery Z-scores ($p = 0.02$). The authors concluded that serum BNP and echocardiographic evaluation performed at the time of BPD diagnosis could detect patients at increased risk of late PH. Moreover, these researchers stated that large, prospective studies are needed to further address this question.

Furthermore, an UpToDate review on "Pulmonary hypertension associated with bronchopulmonary dysplasia" (Collaco and McGrath-Morrow, 2019) states that "Measurements of brain natriuretic peptide (BNP) or N-terminal pro-BNP (NT-proBNP) serum levels are not sufficient to diagnose or exclude PH ... Although studies are limited, serial measurements of serum brain natriuretic peptide (BNP) or N-terminal pro-BNP (NT-proBNP) may also be used to monitor the patient's clinical course and response to treatment". This approach -- serial measurements of serum BNP or NT-proBNP for monitoring patient's clinical course and response to treatment -- is not mentioned in the "Summary and Recommendations" section.

Kingrey et al (2023) noted that patients suffering from PAH demand frequent assessment to keep pace with a dynamic and sometimes rapidly progressive disease course. To improve the understanding of patient monitoring, these researchers carried out a survey of PH providers to establish real-world practice patterns. They assessed the type and frequency of patient assessment methods employed by expert PH providers following PAH diagnosis. A descriptive cross-sectional survey of PH providers across the U.S. was employed to evaluate provider practices. Between September 14, 2017 to October 17, 2017, a survey was distributed electronically to PH experts assessing follow-up frequency and testing evaluation of patients with PAH. A total of 40 (11.4 %) providers completed the survey, representing cardiologists, pulmonologists, and advanced practice providers at centers who cared for an average of 95 patients per year with PAH. Follow-up testing and clinic evaluation was influenced by severity of patient illness. Frequency of re-assessment of clinic follow-up, 6-min walk test (6MWT), echocardiogram, BNP, and right heart catheterization in various clinical scenarios all reflected disparate practice. The authors concluded that current clinical practice patterns in the monitoring of patients with PAH are variable and do not necessarily reflect guideline-based practices, suggesting the need for further research and improved guidelines on the frequency of follow-up and repeat testing.

As a Biomarker for Hypertensive Disorders of Pregnancy

Okwor and colleagues (2020) stated that hypertensive disorders of pregnancy associated with potentially fatal outcomes are common obstetrics occurrences. Early diagnosis, management and prediction of outcomes are challenges to be surmounted especially in developing countries. Biomarkers are emerging as useful tools for diagnosis and prognostication in varying health conditions. Elevated levels of serum copeptin BNP are associated with adverse perinatal outcomes and may serve as potential biomarkers utilized

during routine ante-natal care. In a case-control study, these researchers examined the level and clinical value of copeptin and BNP as biomarkers of hypertensive disorders of pregnancy among Nigerian pregnant women. This trial comprised 156 consenting pregnant women equally grouped into those with chronic hypertension (CH), gestational hypertension (GH), and pre-eclampsia (PE) as cases and normotensives as controls. Pregnant women were recruited from the ante-natal clinic, University College Hospital, Nigeria. Blood pressures (BPs) were measured and blood (10 ml) was drawn from patients, serum and plasma obtained accordingly while other data were collected using interviewer administered questionnaire and medical records. Serum copeptin and plasma BNP levels were measured using enzyme-linked immunosorbent assay (ELISA). Data was analyzed with SPSS version 20.0 and statistical significance was set at $p < 0.05$. The mean levels of systolic BP (SBP) and diastolic BP (DBP) were significantly higher in CH (155.41 ± 2.14 ; 102.36 ± 2.0 mmHg), GH (150.49 ± 0.82 ; 98.67 ± 0.56 mmHg), and PE (153.92 ± 1.47 ; 98.92 ± 0.61 mmHg), compared to controls (101.85 ± 1.9 ; 66.77 ± 1.24 mmHg). Mean serum copeptin and plasma BNP were significantly higher in women with GH (21.25 ± 1.31 pmol/L; 223.05 ± 14.95 pg/ml) and PE (22.47 ± 1.01 pmol/L; 253.99 ± 17.69 pg/ml) compared with controls (9.05 ± 1.01 pmol/L; 48.63 ± 2.50 pg/ml) ($p < 0.05$). There was no significant difference in the mean levels of copeptin and BNP in CH compared with controls ($p > 0.05$). The ROC curve for copeptin gave an AUC of 0.829 ($p = 0.000$) with a cut-off value of 10.15 pmol/L while the AUC for BNP was 0.902 ($p = 0.000$) with a cut-off value of 50.81 pg/ml. The authors concluded that serum copeptin and plasma BNP levels were significantly higher in GH and PE and may be used as markers of hypertensive disorders of pregnancy among pregnant women.

As a Cardiac Biomarker for Friedreich Ataxia

Legrand and colleagues (2020) noted that Friedreich's ataxia (FA) is a rare autosomal recessive mitochondrial disease resulting of a triplet repeat expansion guanine-adenine-adenine (GAA) in the frataxin (FXN) gene, exhibiting progressive cerebellar ataxia, diabetes and cardiomyopathy. These researchers examined the relationship between cardiac biomarkers, serum NT-proBNP, and serum cardiac high-sensitivity troponin (hsTnT) concentrations, and the extent of genetic abnormality and cardiac parameters. Between 2013 and 2015, a total of 85 consecutive genetically confirmed FA adult patients were prospectively evaluated by measuring plasma hsTnT and NT-proBNP concentrations, electrocardiogram, and echocardiography. The 85 FA patients (49 % women) with a mean age of 39 ± 12 years, a mean disease onset of 17 ± 11 years had a mean SARA (Scale for the Assessment and Rating of Ataxia) score of 26 ± 10 . The median hsTnT concentration was 10 ng/L (3 to 85 ng/L) and 34 % had a significant elevated hsTnT of greater than or equal to 14 ng/L. Increased septal wall thickness was associated with increased hsTnT plasma levels ($p < 0.001$). The median NT-proBNP concentration was 31 ng/L (5 to 775 ng/L) and 14 % had significant elevated NT-proBNP of greater than or equal to 125 ng/L. Markers of increased left ventricular filling pressure (trans-mitral E/A and lateral E/E' ratio) were associated with increased NT-proBNP plasma levels ($p = 0.01$ and $p = 0.01$). Length of GAA or the SARA score were not associated with hsTnT or NT-proBNP plasma levels. The authors concluded that cardiac biomarkers such as NT-proBNP and hsTnT plasma levels are easily available and could be included in the routine cardiac evaluation of FA patients, in order to have individual reference values for the disease. In FA, hsTnT could be proposed as a marker of myocardial injury and cardiac involvement, and NT-proBNP would remain a marker of left ventricular filling pressure. Serial measurements are needed to characterize the temporal course of the 2 biomarkers and their relations with the evolution of echocardiographic parameters and the underlying cardiac disease. These researchers stated that further longitudinal studies are needed to evaluate the prognostic value of these biomarkers in FA.

The authors stated that these findings concerned mainly adult FA patients recruited in a single center; therefore, these results could only be generalized to adult patients with FA. It would be of great interest to compare their adult population to pediatric subjects with more severe cardiac disease. Additional drawbacks included the small patient sample size due to the relative rarity of this disease. They stated that further larger collaborative studies are needed to confirm the range of plasma levels of NT-proBNP and hsTnT in FA patients and to define the place of these biomarkers in the management of FA patients.

Prediction of Short-Term Mortality in Individuals with Sepsis

Vallabhajosyula and colleagues (2020) noted that data are conflicting regarding the

optimal cut-offs of BNP and NT-proBNP for prediction of short-term mortality in individuals with sepsis. These researchers conducted a comprehensive search of several data-bases (Medline, Embase, Cochrane Central Register of Controlled Trials, Cochrane Database of Systematic Reviews, and Scopus) for English-language reports of studies evaluating adult patients with sepsis, severe sepsis, and septic shock with BNP/NT-proBNP levels and short-term mortality (ICU, in-hospital, 28-day, or 30-day) published from January 1, 2000, to September 5, 2017. The average values in survivors and non-survivors were used to estimate the receiver operating characteristic curve (ROC) using a parametric regression model. A total of 35 observational studies (3,508 subjects) were included (median age of 51 to 75 years; 12 % to 74 % men; cumulative mortality, 34.2 %). A BNP of 622 pg/ml had the greatest discrimination for mortality (sensitivity, 0.695 [95 % CI: 0.659 to 0.729]; specificity, 0.907 [95 % CI: 0.810 to 1.003]; area under the ROC, 0.766 [95 % CI: 0.734 to 0.797]). An NT-proBNP of 4,000 pg/ml had the greatest discrimination for mortality (sensitivity, 0.728 [95 % CI: 0.703 to 0.753]; specificity, 0.789 [95 % CI: 0.710 to 0.867]; area under the ROC, 0.787 [95 % CI: 0.766 to 0.809]). In pre-specified subgroup analyses, identified BNP/NT-proBNP cutoffs had higher discrimination if specimens were obtained 24 hours or less after admission, in patients with severe sepsis/septic shock, in patients enrolled after 2010, and in studies performed in the U.S. and Europe. There was inconsistent adjustment for renal function. The authors concluded that in this hypothesis-generating analysis, BNP and NT-proBNP cut-offs of 622 pg/ml and 4,000 pg/ml optimally predicted short-term mortality in patients with sepsis. The applicability of these results was limited by the heterogeneity of included patient populations. These researchers stated that further dedicated research into the incorporation of these biomarkers into prognostic models and structured evaluation of cardiovascular dysfunction in patients with sepsis are needed to understand the clinical implications of these findings.

NT-proBNP for Prediction of Acute Kidney Injury After Non-Cardiac Surgery

Zhao and colleagues (2021) stated that acute kidney injury (AKI) is associated with poor outcomes after non-cardiac surgery. Whether pre-operative NT-proBNP predicts AKI following non-cardiac surgery is unclear. In a retrospective, cohort study, these investigators examined the predictive role of pre-operative NT-proBNP on post-operative AKI. Adult patients who had a serum creatinine and NT-proBNP measurement within 30 pre-operative days and at least 1 serum creatinine measurement within 7 days after non-cardiac surgery between February 2008 and May 2018 were identified. The primary outcome was post-operative AKI, defined by the kidney disease: improving global outcomes creatinine criteria. In all, 6.1 % (444 of 7,248) of patients developed AKI within 1 week after surgery. Pre-operative NT-proBNP was an independent predictor of AKI after adjustment for clinical variables (OR comparing top to bottom quintiles 2.29, 95 % CI: 1.47 to 3.65, $p < 0.001$ for trend; OR per 1-unit increment in natural log transformed NT-proBNP 1.27, 95 % CI: 1.16 to 1.39). Compared with clinical variables alone, the addition of NT-proBNP improved model fit, modestly improved the discrimination (change in area under the curve from 0.764 to 0.773, $p = 0.005$) and re-classification (continuous net re-classification improvement 0.210, 95 % CI: 0.111 to 0.308, improved integrated discrimination 0.0044, 95 % CI: 0.0016 to 0.0072) of AKI and non-AKI cases, and achieved higher net benefit in decision curve analysis. The authors concluded that pre-operative NT-proBNP concentrations provided predictive information for AKI in a cohort of patients undergoing non-cardiac surgery, independent of and incremental to conventional risk factors. Moreover, these researchers stated that prospective studies are needed to confirm these findings and examined its clinical impact.

NT-proBNP for Prediction of Cardiovascular Complications After Bariatric Surgery

van Veldhuisen and colleagues (2021) noted that obesity is associated with cardiovascular (CV) risk factors and diseases. Because bariatric surgery is increasingly being carried out in relatively elderly patients, a risk for pre- and post-operative CV complications exists. In a single-center, cohort study, these researchers examined the value of plasma NT-proBNP as a CV screening tool. Between June 2019 and January 2020, all consecutive bariatric patients 50 years and older underwent pre-operative NT-proBNP evaluation to screen for CV disease. Patients with elevated NT-proBNP (greater than or equal to 125 pg/ml) were referred for further cardiac evaluation, including electrocardiography (ECG) and echocardiography. A total of 310 consecutive patients (median age of 56 years; 79 % female; body mass index [BMI] = 43 ± 6.5 kg/m²) were included in this trial. A history of CV disease was present in 21 % of patients, mainly atrial fibrillation (AF; 7 %) and coronary artery disease (CAD; 10 %); 72 patients (23 %) had elevated NT-proBNP levels,

and 67 of them underwent further cardiac work-up. Of these 67 patients, ECG showed AF in 7 patients (10 %). On echocardiography, 3 patients had left ventricular ejection fraction (LVEF) of less than 40 %, 9 patients had LVEF of 40 % to 49 %, and 13 patients had LVEF of greater than or equal to 50 % with structural and/or functional re-modeling.

In 2 patients, elevated NT-proBNP prompted work-up resulting in a diagnosis of CAD and consequent percutaneous coronary intervention (PCI) in 1 patient. The authors concluded that elevated NT-proBNP levels were present in 23 % of patients 50 years and older undergoing bariatric surgery. In 37 % of them, there was ECG evidence for structural and/or functional re-modeling. Moreover, these researchers stated that further studies are needed to examine if these preliminary results warrant routine application of NT-proBNP to identify patients at risk for CV complications following bariatric surgery.

The authors stated that this study had several drawbacks. First, this was a single-center study with a relatively small sample size; thus, the findings may not provide conclusive evidence whether cardiac screening is beneficial in terms of reducing CV morbidity and mortality. Second, patients with normal NT-proBNP levels were not referred for ECG or echocardiography; thus, the presence of CV disease was unknown in these patients. However, a normal BNP or NT-proBNP made it very unlikely that a patient has CV disease, especially HF. It should also be noted that NT-proBNP is a stronger predictor for HF than CAD and stroke, although these researchers did identify 2 patients in their cohort who needed intervention for CAD while using NT-proBNP. Third, the objective of this trial was to examine the value of NT-proBNP as a screening tool for CV disease and therefore was not powered to examine a potential association with post-operative CV outcome. Fourth, NT-proBNP has an inverse relation with BMI, which means that patients with potential CV disease could have false-negative outcome of cardiac screening with NT-proBNP. This implied that the reported 23 % of patients with elevated NT-proBNP in this study was probably an under-estimation of the actual number of patients with CV disease.

Fifth, use of NT-proBNP as a single screening tool might be less predictive for CV disease than a prediction model that combines NT-proBNP with levels of additional laboratory measurements (e.g., troponins or highly sensitive C-reactive protein [hs-CRP]) and presence of co-morbidities such as diabetes. However, these investigators aimed to examine NT-proBNP as a simple and stand-alone diagnostic tool; thus, they did not add other parameters to the decision whether or not patients should be referred for cardiac work-up. Last, it was likely that this cohort of patients was slightly different than the general obese population. General practitioners may be reluctant to refer a patient for bariatric surgery if they have CV disease such as CHF or recent myocardial infarction (MI) because of a higher risk of fatal complications following bariatric surgery.

Diagnosis of Systemic Sclerosis Heart Involvement

Ross et al (2021) noted that systemic sclerosis (SSc) heart involvement (SHI) is a leading cause of SSc-associated mortality and once clinically overt, carries a very poor prognosis. There remain no established diagnostic criteria for SHI. In a systematic review, these investigators examined the literature regarding the role of cardiac troponin (cTn) and BNP or NT-proBNP in the diagnosis of SHI. They carried out a comprehensive search of the Medline (Ovid), Embase and PubMed databases to identify adult human studies of at least 10 SSc patients with a primary focus of SHI that included data on cTn and BNP or NT-proBNP results. Only cohort studies and case-controlled studies were identified; and the quality of the evidence presented in each study was assessed according to the Newcastle-Ottawa Quality Assessment Scale. Of the 2,742 studies identified by the database search, 12 articles met the study inclusion criteria; 3 out of the 4 studies evaluating SHI using cardiac MRI found no association between cardiac biomarkers and imaging changes. By comparison, echocardiographic abnormalities, cardiac arrhythmias and congestive HF were more likely to be associated with elevated cardiac biomarkers. Comparison of results between studies was limited by the highly heterogeneous definitions of SHI and inclusion criteria employed across studies. The authors concluded that there are insufficient data to draw definitive conclusions regarding the role of cTn and BNP / NT-proBNP in the diagnosis of SHI. These researchers stated that currently available literature suggested that cardiac biomarkers may have some role, in conjunction with other diagnostic modalities, in identifying SHI; however, this remains a much-needed area of clinical research.

Prognostic Biomarker of Weaning Outcome from Mechanical Ventilation

Deschamp et al (2020) noted that predicting successful liberation from mechanical ventilation (MV) in critically ill patients is challenging; BNP has been proposed to aid

in guiding decision-making for readiness to liberate from MV following a spontaneous breathing trial (SBT). These researchers carried out a systematic review and meta-analysis of randomized and prospective observational studies that measured BNP levels at the time of SBT in patients receiving MV. The primary endpoint was successful liberation from MV (absence of re-intubation or non-invasive ventilation at 48 hours). Statistical analyses included bi-variate and Moses-Littenberg models and DerSimonian-Laird pooling of areas under ROC curve (AUROC). A total of 731 articles were screened; 18 adult and 2 pediatric studies were fulfilled pre-specified eligibility. The measure of the relative variation of BNP during SBT (Δ BNP%) after exclusion of SBT failure by clinical criteria in adults yielded a sensitivity and specificity of 0.889 [0.831 to 0.929] and 0.828 [0.730 to 0.896] for successful liberation from MV, respectively, with a pooled AUROC of 0.92 [0.88 to 0.97]. The pooled AUROC for any method of analysis for absolute variation of BNP (Δ BNP), pre-SBT BNP, and post-SBT BNP were 0.89 [0.83 to 0.95], 0.77 [0.63 to 0.91], and 0.85 [0.80 to 0.90], respectively. The authors concluded that the relative change in BNP during a SBT has potential value as an incremental tool following successful SBT to predict successful liberation from MV in adults. Moreover, these researchers stated that here is insufficient data to support the use of BNP in children or as an alternate test to clinical indices of SBT, or the use of Δ BNP, BNP-pre, and BNP-post as an alternate or incremental test. They noted that studies comparing the best use of Δ BNP% either as an alternative or incremental tool to clinical indices during SBT as well as prospective validation of a specific threshold represent the next step in research. There is paucity of data in pediatric cases that limited any conclusion.

Liu et al (2021) stated that MV is an important treatment for critically ill patients.

Physicians usually conduct a SBT to examine if the patients can be weaned from MV; however, approximately 17 % of the patients who passed the SBT still require respiratory support. Cardiac dysfunction is an important cause of weaning failure. The use of BNP or NT-proBNP is a simple method to evaluate cardiac function. In a systematic review, these investigators examined the use of BNP or NT-proBNP as predictors of weaning from MV. Data sources included PubMed (1950 to December 2020), Cochrane, and Embase (1974 to December 2020), and some Chinese databases for additional articles (China Biology Medicine (CBM), China Science and Technology Journal Database (CSTJ), and Wanfang Data and China National Knowledge Infrastructure (CNKI)). These researchers systematically searched observation studies examining the predictive value of BNP or NT-proBNP in weaning outcome of patients with MV. Two independent reviewers extracted data; the differences were resolved through consultation. These investigators included 18 articles with 1,416 patients and extracted 6 index tests with pooled sensitivity and specificity for each index test. For the BNP change rate predicting weaning success, the pooled sensitivity was 89 % (83 % to 94 %) and the pooled specificity was 82 % (72 % to 89 %) with the highest pooled AUC of 0.9511. The authors concluded that the BNP change rate was a reliable predictor of weaning outcome from MV. Moreover, these researchers stated that more diagnostic randomized controlled trials (RCTs) are needed to establish the best use of this diagnostic indicator.

Wu et al (2021) noted that cardiovascular dysfunction has been reported as an important mechanism of weaning failure, and recent data suggested that elevated BNP levels is associated with an increased risk of weaning failure. In a meta-analysis, these investigators examined the correlation between elevated plasma BNP levels and weaning failure in critically ill patients subject to MV. They carried out a systematic search in Cochrane Library, Embase, PubMed and Web of Science up to September 25, 2019.

Standard mean differences (SMD) and corresponding 95 CIs of the BNP levels were calculated for each study. A total of 9 studies with a total number of 589 were included in the final meta-analysis. The results showed that elevated BNP levels were significantly associated with the risk of weaning failure (SMD: 0.76, 95 % CI: 0.47 to 1.05, $p < 0.00001$). The finding was consistent with the BNP measured before (SMD: 0.68, 95 % CI: 0.26 to 1.11, $p = 0.002$) or at the end of SBT (SMD: 0.85, 95 % CI: 0.52 to 1.18, $p < 0.00001$). The authors concluded that the findings of this meta-analysis showed that measurement of plasma BNP levels is a promising tool for early identification of patients having difficulty in weaning of MV. Moreover, these researchers stated that larger and more adequately powered cohort studies are needed to identify the assay standardization, the optimal cut-off point, and the predictive value of BNP levels for weaning outcome in the future.

The authors stated that this meta-analysis had several drawbacks. First, BNP is a valuable biomarker in response to volume overload and LV dysfunction; however, elevated plasma BNP levels are not specific to HF, and circulating levels may be affected by many factors (e.g., age, sex, obesity and renal function), which may influence the release and

clearance of the cardiac hormone. Second, in recent years, BNP measurement assays have become conveniently available; and they were used in the various clinical settings.

These include both point-of-care tests and high-throughput automated platforms.

However, the lack of assay standardization made it difficult to analyze equivalent currently; thus, clinicians should carefully evaluate results acquired by different laboratories using different analytical approaches. Third, an SBT can be performed with patients breathing without any ventilatory support or with minimal pressure support. Not all SBTs generate a similar increase in pulmonary arterial occlusion pressure (PAOP). Experimental studies have shown that compared with T-piece trial, the use of any level of pressure support decreased the inspiratory load significantly, as such, the most noted decrease in PAOP. Accordingly, the T-piece trial was the most appropriate choice to unmask the development of LV heart failure during spontaneous breathing in patients with cardiac dysfunction. Therefore, when considering the most reasonable approach to use in a weaning trial, this finding should be taken into account. Fourth, these researchers were unable to identify a reliable cut-off point for BNP tests because they did not have the raw data to map out ROC curves. Furthermore, the sample size of this study was relatively small; and the accuracy of these findings may be affected.

Diagnostic Value of Cardiac Natriuretic Peptide on Pulmonary Hypertension in Systemic Sclerosis

Zhang et al (2022) noted that pulmonary arterial hypertension (PAH) is a major cause of morbidity and mortality in SSc. Many risk factors and predictors of outcomes have been identified in these patients; BNP and NT-proBNP serum levels are often elevated in SSc patients with early PAH. In a systematic review and meta-analysis, these researches examined the diagnostic value of BNP/NT-proBNP in SSc secondary PAH (SSc-PAH). They carried out a systematic search via PubMed, Embase, and Cochrane Library databases up to January 30, 2021. Stata 16.0 (Stata Corp, College Station, TX) was used to perform the meta-analysis. A total of 9 studies involving 220 SSc-PAH patients and 259 non-SSc-PAH controls were included. The values of sensitivity and specificity using BNP and NT-ProBNP as diagnostic tools were pooled in the diagnostic meta-analysis. The overall performance of BNP/NT-ProBNP detection was as follows: pooled sensitivity, 0.67 (95 % CI: 0.52 to 0.79); pooled specificity, 0.84 (95 % CI: 0.75 to 0.91); pooled positive likelihood ratio (PLR), 4.3 (95 % CI: 3 to 6.1); and pooled negative likelihood ratio (NLR), 0.39 (95 % CI: 0.28 to 0.55). The subgroup analysis showed similar results. Funnel plots indicated that there was no evidence for publication bias. The authors concluded that these findings revealed that NT-proBNP has certain diagnostic value for PAH due to its better specificity and moderate sensitivity; however, its clinical application value remains sub-optimal and could not be a stand-alone decision-making diagnostic tool of SSc-PAH.

Screening for Left Ventricular Systolic Dysfunction

Goyder et al (2023) stated that HF is a global health burden and new strategies to achieve timely diagnosis and early intervention are urgently needed. Natriuretic peptide (NP) testing can be used to screen for left ventricular systolic dysfunction (LVSD); however, evidence on test performance is mixed, and international HF guidelines differ in their recommendations. In a systematic review and meta-analysis, these investigators examined the evidence on diagnostic accuracy of NP screening for LVSD in general and high-risk community populations and estimated optimal screening thresholds. They searched relevant databases up to August 2020 for studies with a screened community population of over 100 adults reporting NP performance to diagnose LVSD. Study inclusion, quality assessment, and data extraction were carried out independently and in duplicate. Diagnostic test meta-analysis used hierarchical summary ROC curves to obtain estimates of pooled accuracy to detect LVSD, with optimal thresholds obtained to maximize the sum of sensitivity and specificity. A total of 24 studies were identified, involving 26,565 subjects: 8 studies in high-risk populations (at least 1 cardiovascular risk factor), 12 studies in general populations, and 4 in both high-risk and general populations combined. For detecting LVSD in screened high-risk populations with NNT-proBNP, the pooled sensitivity was 0.87 (95 % CI: 0.73 to 0.94) and specificity 0.84 (95 % CI: 0.55 to 0.96); for BNP, sensitivity was 0.75 (95 % CI: 0.65 to 0.83) and specificity 0.78 (95 % CI: 0.72-0.84). Heterogeneity between studies was high with variations in positivity threshold. Due to a paucity of high-risk studies that examined NP performance at multiple thresholds, it was not possible to calculate optimal thresholds for LVSD screening in high-risk populations alone. To provide an indication of where the positivity threshold might lie, the pooled accuracy for LVSD screening in

high-risk and general community populations were combined and gave an optimal cut-off of 311 pg/ml (sensitivity 0.74 (95 % CI: 0.53 to 0.88), specificity 0.85 (95 % CI: 0.68 to 0.93) for NT-proBNP and 49 pg/ml (sensitivity 0.68 (95 % CI: 0.45 to 0.85), specificity 0.81 (0.67 to 0.90) for BNP. The authors concluded that these findings suggested that in high-risk community populations NP screening may accurately detect LVSD, potentially providing an important opportunity for diagnosis and early intervention. Moreover, these researchers stated that this study highlighted an urgent need for further prospective studies, as well as an individual participant data meta-analysis, to more precisely examine diagnostic accuracy and identify optimal screening thresholds in specifically defined community-based populations as well as to further examine the impact of NP screening on both general and high-risk populations to inform future guideline recommendations.

Screening of Heart Failure in Patients with Diabetes Mellitus

In a prospective study, Resl et al (2016) hypothesized that biomarkers representing different pathophysiological pathways of atherosclerosis, namely NT-proBNP, growth differentiation factor 15 (GDF-15), and high-sensitive troponin T (hs-TnT) could enhance cardiovascular risk prediction in patients with type 2 diabetes mellitus (T2DM). This study included 746 patients with T2DM, who were followed-up for 60 months. The primary endpoint was defined as unplanned hospitalization for cardiovascular disease (CVD) or death. The prognostic performance of the biomarkers of interest (GDF-15 in comparison with NT-proBNP and hs-TnT) was examined in univariate as well as in step-wise Cox regression models. Hazard ratios (HRs) were presented per standard unit increase. The primary endpoint was registered in 171 patients (22.9 %). In univariate Cox regression models, GDF-15 as well as hs-TnT provided significant prognostic information. Even after adjusting for established CVD risk factors, GDF-15, hs-TnT, and NT-proBNP remained strong independent predictors of the endpoint (logGDF-15: HR 1.37, $p < 0.01$, CI: 1.12 to 1.68; loghs-TnT: HR 1.43, $p < 0.01$, CI: 1.13 to 1.82; logNT-proBNP: HR 1.45, $p < 0.01$, CI: 1.26 to 1.66). The number of elevated markers showed a strong complementarity to predict future long-term risk. Adding hs-TnT and GDF-15 to a zero model already including NT-proBNP resulted in a net re-classification improvement (NRI) of 33.6 % (CI: 16.0 % to 50.8 %, NRI for patients with event: 11.1 % CI: -4.7 % to 26.6 %, for patients without event: 22.5 % CI 13.6 % to 30.5 %). The authors concluded that GDF-15 and hs-TnT are strong independent CVD biomarkers augmenting the predictive value of NT-proBNP in patients with T2DM.

Ontario Health (Quality)' technology assessment on "Use of B-type natriuretic peptide (BNP) and N-terminal proBNP (NT-proBNP) as diagnostic tests in adults with suspected heart failure" (2021) noted that heart failure (HF) is a complex clinical syndrome in which abnormal cardiac function increases the risk of or results in clinical symptoms and signs of reduced cardiac output and/or pulmonary or systemic congestion. The syndrome can be acute or chronic and often develops after other conditions, such as hypertension, coronary artery disease (CAD), or DM, or behavioral factors such as heavy alcohol use, have damaged or weakened the heart.

In a systematic review and meta-analysis, Ramzi (2023) examined the available evidence on the prognostic performance of NT-proBNP in predicting cardiovascular events, cardiovascular-related mortality, as well as all-cause mortality in patients with T2DM. These investigators carried out searches in Medline, Embase, Scopus, and Web of Science databases before August 1, 2021; the data were recorded as adjusted HR. An increase in NT-proBNP increases the risk of cardiovascular events (HR = 1.63), cardiovascular mortality (HR = 1.86), and all-cause mortality (HR = 1.54). Seemingly, the best cut-offs for predicting cardiovascular events (HR = 2.30) and cardiovascular mortality (HR = 3.77) were levels greater than 100 pg/ml. The best cut-off of NT-proBNP in predicting all-cause mortality was levels greater than 225 pg/ml (HR = 4.72). The author concluded that a moderate level of evidence showed that NT-proBNP serum levels could predict future cardiovascular events, cardiovascular mortality, and all-cause mortality; therefore, it could be used as risk stratification for T2DM.

In a systematic review and meta-analysis, Ahmad et al (2024) identified potentially novel prognostic factors that may improve cardiovascular disease (CVD) risk prediction in T2DM.

Out of 9,380 studies identified, 416 studies met inclusion criteria. Outcomes were reported for 321 biomarker studies, 48 genetic marker studies, and 47 risk score/model studies. Out of all evaluated biomarkers, only 13 showed improvement in prediction performance. Results of pooled meta-analyses, non-pooled analyses, and assessments of improvement in prediction performance and risk of bias, yielded the highest predictive

utility for NT-proBNP (high-evidence), TnT (moderate-evidence), triglyceride-glucose (TyG) index (moderate-evidence), Genetic Risk Score for Coronary Heart Disease (GRS-CHD) (moderate-evidence); moderate predictive utility for coronary computed tomography angiography (CCTA; low-evidence), single-photon emission computed tomography (SPECT; low-evidence), pulse wave velocity (moderate-evidence); and low predictive utility for C-reactive protein (CRP; moderate-evidence), coronary artery calcium score (low-evidence), galectin-3 (low-evidence), troponin-I (low-evidence), carotid plaque (low-evidence), and GDF-15 (low-evidence). Risk scores showed modest discrimination, with lower performance in populations different from the original development cohort. The authors concluded that despite high interest in this topic, very few studies conducted rigorous analyses to demonstrate incremental predictive utility beyond established CVD risk factors for T2DM.

The most promising markers identified were NT-proBNP, TnT, TyG, and GRS-CHD, with the highest strength of evidence for NT-proBNP.

The American Diabetes Association (ADA)'s guidelines on "Cardiovascular disease and risk management: Standards of care in diabetes" (2024) states that "Adults with diabetes are at increased risk for the development of asymptomatic cardiac structural or functional abnormalities (stage B heart failure) or symptomatic (stage C) heart failure. Consider screening adults with diabetes by measuring a natriuretic peptide (B-type natriuretic peptide [BNP] or N-terminal pro-BNP [NT-proBNP]) to facilitate prevention of stage C heart failure. Level of Evidence = B (Supportive evidence from well-conducted cohort studies)".

N-Terminal Pro B-Type Natriuretic Peptide for Monitoring the Effectiveness of Therapy for Patients with Congestive Heart Failure

Januzzi et al (2020) stated that canagliflozin lowers cardiovascular events including hospitalization for HF (HHF) in patients with T2DM and cardiovascular risk. Elevated NT-proBNP concentrations are associated with HF diagnosis and predict cardiovascular risk. These researchers measure NT-proBNP in CANVAS (Canagliflozin Cardiovascular Assessment Study) participants; they ascertained associations between baseline NT-proBNP and cardiovascular, renal, as well as mortality outcomes and intervention-associated changes. Of the 4,330 participants in the CANVAS Trial, NT-proBNP was measured in 3,587, 2,918, and 995 participants at baseline, 1 year, and 6 years, respectively. The median baseline NT-proBNP concentration was 91 pg/ml, and 39.3 % had NT-proBNP of 125 pg/ml or higher. NT-proBNP was higher in those with investigator-reported HF (13 % of participants at baseline) versus those without (187 pg/ml versus 81 pg/ml), with substantial overlap between groups. By 1 year, NT-proBNP increased with placebo, whereas canagliflozin reduced NT-proBNP by 11 % (geometric mean ratio for canagliflozin versus placebo = 0.89 [95 % CI: 0.84 to 0.94]; $p < 0.001$). Lower NT-proBNP with canagliflozin was also observed at 6 years ($p = 0.004$). In adjusted models, baseline NT-proBNP of 125 pg/ml or higher was prognostic for incident HHF (HR: 5.40; 95 % CI: 2.67 to 10.9), HHF/ cardiovascular death (HR: 3.52; 95 % CI: 2.38 to 5.20), and all-cause death (HR: 2.53; 95 % CI: 1.78 to 3.61). Mediation analyses suggested that 10.4 % of the effects of canagliflozin on HHF were reflected in NT-proBNP lowering. The authors concluded that a substantial percentage of patients in the CANVAS Trial had elevated NT-proBNP values. Canagliflozin reduced NT-proBNP concentrations versus placebo; however, reduction in NT-proBNP explained only a small proportion of the benefit of canagliflozin on HF events.

Tanaka et al (2021) noted that sodium-glucose co-transporter 2 (SGLT2) inhibitors reduce the risk of a deterioration in HF and mortality in patients with a broad range of cardiovascular risks. Recent guidelines recommended considering the use of SGLT2 inhibitors in patients with T2DM and HF, irrespective of their glycemic control status and background use of other glucose-lowering agents including metformin. However, only a small number of studies have examined if the effects of SGLT2 inhibitor in these patients differ by the concomitant use of other glucose-lowering agents. In a post-hoc analysis of the CANDLER Trial, an investigator-initiated, open-label, multi-center RCT, these researchers examined the effect of 24 weeks of treatment with canagliflozin, relative to glimepiride, on NT-proBNP concentration in patients with T2DM and clinically stable chronic HF. The effect of canagliflozin on NT-proBNP concentration was evaluated in the patients according to their baseline use of other glucose-lowering agents. Almost all patients in the CANDLER Trial presented as clinically stable (NYHA class I to II), with approximately 70 % of participants having HF with a preserved EF phenotype (defined as a LVEF of 50 % or greater) at baseline. Of the 233 patients randomized to either canagliflozin (100 mg daily) or glimepiride (starting dose 0.5 mg daily), 85 (36.5 %) had not been taking any glucose-lowering agents at baseline (naïve). Of the 148 patients who

had been taking at least 1 glucose-lowering agent at baseline (non-naïve), 44 (29.7 %) and 127 (85.8 %) had received metformin or a dipeptidyl dipeptidase-4 (DPP-4) inhibitor, respectively. The group ratio (canagliflozin versus glimepiride) of proportional changes in the geometric means of NT-proBNP concentration was 0.95 (95 % CI: 0.76 to 1.18, $p = 0.618$) for the naïve subgroup, 0.92 (95 % CI: 0.79 to 1.07, $p = 0.288$) for the non-naïve subgroup, 0.90 (95 % CI: 0.68 to 1.20, $p = 0.473$) for the metformin-user subgroup, and 0.91 (95 % CI: 0.77 to 1.08, $p = 0.271$) for the DPP-4 inhibitor-user subgroup. No heterogeneity in the effect of canagliflozin, relative to glimepiride, on NT-proBNP concentration was observed in the non-naïve subgroups compared to that in the naïve subgroup. The authors concluded that the impact of canagliflozin treatment on NT-proBNP concentration appeared to be independent of the background use of diabetes therapy in the patient population examined.

Heidenreich et al (2022) noted that the “2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure” replaced the “2013 ACCF/AHA Guideline for the Management of Heart Failure” and the “2017 ACC/AHA/HFSA Focused Update of the 2013 ACCF/AHA Guideline for the Management of Heart Failure”. The 2022 guideline was intended to provide patient-centric recommendations for clinicians to prevent, diagnose, and manage patients with HF. These investigators carried out a comprehensive literature search from May 2020 to December 2020, entailing studies, reviews, and other evidence performed on human subjects that were published in English from Medline (PubMed), Embase, the Cochrane Collaboration, the Agency for Healthcare Research and Quality (AHRQ), and other relevant data-bases. Additional relevant clinical trials and research studies, published through September 2021, were also considered. This guideline was harmonized with other AHA/ACC guidelines published through December 2021. Heart failure remains a leading cause of morbidity and mortality worldwide. The 2022 HF guideline provided recommendations based on contemporary evidence for the treatment of these patients. The recommendations presented an evidence-based approach to managing patients with HF, with the intent to improve quality of care and align with patients’ interests. Many recommendations from the earlier HF guidelines have been updated with new evidence, and new recommendations have been created when supported by published data. Value statements were provided for certain treatments with high-quality published economic analyses. One of the “10 Take Home Messages” was that available evidence supporting increased filling pressures is important for the diagnosis of HF if the LVEF is greater than 40 %. Furthermore, evidence for increased filling pressures can be obtained from non-invasive (e.g., natriuretic peptide, diastolic function on imaging) or invasive testing (e.g., hemodynamic measurement).

In an update of the 2021 European Society of Cardiology (ESC) guidelines for the diagnosis and treatment of acute and chronic HF, McDonagh et al (2023) stated that the Task Force did not specify NT-proBNP thresholds for treatment, consistent with recommendations for other therapies in the original 2021 ESC HF guidelines; however, they noted that, in the diagnostic algorithm for HF in the 2021 ESC HF guidelines, increased concentrations of natriuretic peptides are usually implicit to that diagnosis.

Fuery et al (2024) stated that although clinical studies have shown the association between a single NT-proBNP measurement and clinical outcomes in chronic HF, the biomarker is often measured serially in clinical practice. These investigators determine the added prognostic value of repeated NT-proBNP measurements compared with single measurements alone for patients with chronic HF. In the GUIDE-IT (Guiding Evidence Based Therapy Using Biomarker Intensified Treatment in Heart Failure) Trial, a total of 894 patients with chronic HF with reduced EF were enrolled at 45 outpatient sites in the U.S. and Canada. Repeated NT-proBNP levels were measured over a 2-year study period. Associations between repeated NT-proBNP measurements and trial end-points were assessed using a joint longitudinal and survival model. After adjustment for baseline co-variates, each doubling of the baseline NT-proBNP level was associated with a HR of 1.17 (95 % CI: 1.08 to 1.28; $p = 0.0003$) for the primary trial end-point of cardiovascular death or HHF. Serial measurements increased the adjusted HR for the primary trial end-point to 1.66 (95 % CI: 1.50 to 1.84; $p < 0.0001$), and a similar increased risk was observed across secondary trial end-points. In joint modeling, an increase in NT-proBNP occurred weeks before the onset of adjudicated events. The authors concluded that repeated NT-proBNP measurements were a strong predictor of outcomes in HF with reduced EF with an increase in concentration occurring well before event onset. These results may support routine NT-proBNP monitoring to assist in clinical decision-making.

N-Terminal Pro B-Type Natriuretic Peptide for Cardiac Dysfunction in Asymptomatic Long-

Term Breast Cancer Survivors

Knol et al (2025) noted that with a growing population of breast cancer (BC) survivors, it is important to acknowledge long-term consequences of BC treatment, including left ventricular systolic dysfunction (LVSD). Although echocardiography (ECG) is a reliable technique to diagnose LVSD, its limited accessibility in primary care poses challenges. These researchers carried out a cross-sectional diagnostic accuracy study among 350 long-term BC survivors (at least 5 years after BC diagnosis), comparing the diagnostic performance of index tests ECG and NT-proBNP to the reference test ECG. LVSD was defined as LVEF of less than 54 % or LVEF less than 50 % on ECG. The median age at time of investigation was 63 years (IQR 57 to 68), with a median follow-up duration since BC diagnosis of 10 years (IQR 7 to 14). An abnormal ECG showed a sensitivity of 63.0 % (IQR 48.7 % to 75.7 %), a corresponding specificity of 51.7 % (IQR 45.8 % to 57.6 %) and a NLR of 0.7 (IQR 0.5 to 1.0) for detecting a LVEF of less than 54 %. An abnormal ECG showed a sensitivity of 75.0 % (IQR 47.6 % to 92.7 %), a corresponding specificity of 50.6 % (IQR 45.1 % to 56.2 %) and a NLR of 0.5 (0.2 to 1.2) for detecting LVSD defined as LVEF of less than 50 %. The area under the curve (AUC) for NT-proBNP was 0.59 (95 % CI: 0.50 to 0.68) for detecting LVEF of less than 54 % and 0.56 (95 % CI: 0.39 to 0.74) for detecting LVEF of less than 50 %. The authors concluded that ECG and NT-proBNP were inadequate diagnostic tools to screen for LVSD among asymptomatic long-term BC survivors.

Serial N-Terminal Pro B-Type Natriuretic Peptide Testing After Bi-Ventricular Repair in Patients with Borderline Hypoplastic Left Ventricle

Osawa et al (2025) stated that bi-ventricular repair (BVR) for patients with borderline hypoplastic left ventricle is challenging, and a predictor of failing BVR has not been established. In a retrospective, non-randomized, single-center study, these investigators examined the usefulness of z-log NT-proBNP in predicting outcomes following BVR. Patients who were diagnosed with borderline left heart hypoplasia and underwent BVR from 2012 to 2022 were included. Serial NT-proBNP values were evaluated using its age-adjusted z-score. The data were collected from the 1st admission to the last follow-up and compared between patients with failing BVR (defined as death and hemodynamic failure) and patients with hemodynamically good bi-ventricular outcomes. A total of 34 patients were included, and 7 patients (21 %) developed adverse outcomes (5 deaths and 2 hemodynamic failures) following BVR. The mean value of z-log NT-proBNP before BVR was not significantly different between patients with failing BVR and those without (2.2 [1.5 to 3.2] versus 3.3 [2.2 to 3.9], $p = 0.200$). However, patients with failing BVR showed a continuous increase in NT-proBNP post-operatively. The value for patients with failing BVR was higher within 7 days after BVR ($p = 0.016$) and at the last follow-up ($p = 0.003$) than those without. Post-operative z-log NT-proBNP and endocardial fibroelastosis at birth were identified as associated factors of failing BVR. The authors concluded that elevated z-log NT-proBNP after BVR of borderline hypoplastic left ventricle appeared to be a useful biomarker associated with poor outcomes.

The authors stated that this trial was limited by its retrospective, non-randomized, and single-center design. The sample size was small ($n = 34$), which may limit the generalizability of the findings. In addition, while these researchers did not carry out formal sample size or power calculations before the study, their analysis included statistical tests and confidence intervals that provided insight into the robustness of the findings. This study included various diagnoses, such as unbalanced atrio-ventricular (AV) septal defect and conotruncal heart malformations (CTMs). The parameters of the echo at birth are dynamic and may be interpreted differently depending on the time of adoption. LV volume and other LV parameters were measured by ECG. It made it difficult to compare these data to other reports, where cardiac MRI was the typical modality used for the calculation of ventricular volumes. Another drawback was the variability in the timing of NT-proBNP measurements, which could affect the interpretation of the results. These researchers stated that standardizing the timing of these measurements in future studies could provide more consistent and reliable data.

Serum N-Terminal Pro-Brain Natriuretic Peptide, D-Dimer, Albumin Combined with T-Cell Subsets in Detecting Coronary Artery Damage in Children with Kawasaki Disease

Wang et al (2025) noted that Kawasaki disease (KD), a self-limiting vasculitis in children, often affecting coronary arteries, which may result in coronary artery dilation, stenosis, or in severe cases, MI. These investigators identified new

approaches for reducing or preventing coronary artery lesions (CAL) in KD patients by analyzing specific serological markers across various pediatric groups. Clinical data were collected from 100 children diagnosed with KD admitted at First Affiliated Hospital of Hebei North University between May 2023 and June 2024. These children were divided into 2 groups based on coronary artery injury status: Occurrence group (n = 31) and non-occurrence group (n = 69). Furthermore, data from 100 children with acute upper respiratory tract infections (URTI) and 100 healthy children who underwent routine physical examination during the same period (Healthy group) were included for comparison. Serum levels of NT-proBNP, D-dimer (D-D), albumin (ALB), and T-cell subsets were measured and compared across groups to examine their clinical utility in diagnosing coronary artery damage in KD. NT-proBNP and D-D levels were highest in KD children with coronary artery injury and lowest in the healthy group, while ALB levels were lowest in KD children with coronary artery injury and highest in the healthy group, with statistically significant differences ($p < 0.001$). Analysis of T-cell subsets showed that cluster of differentiation (CD)3+, CD4+, and CD4+/CD8+ levels were highest in the Healthy group, while CD8+ levels were highest in the Occurrence group, with statistically significant differences ($p < 0.001$). The combined diagnostic model showed an AUC value of 0.885 (95 % CI: 0.829 to 0.941), showing higher specificity and AUC value compared to each marker individually. The authors concluded that the combination of serum NT-proBNP, D-D, ALB, and T-cell subsets offered valuable predictive insights for coronary artery damage in KD children and may serve as an auxiliary diagnostic tool.

These researchers stated that while the findings of this trial provided valuable insights into the early diagnosis of KD with CAL, it has several drawbacks. Notable factors included a relatively small sample size and limitations in laboratory testing methods, which may impact the comprehensiveness and objectivity of the conclusions. These investigators stated that these drawbacks underscored the importance of conducting future studies with increased rigor to enhance the accuracy and reliability of these findings and to minimize the impact of these potential drawbacks on experimental conclusions.

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